

knn

April 17, 2026

Generative AI Use: For the purposes of the assignments, the use of generative AI is subject to the same policies regarding collaboration. Just as with other collaborators, each student must write down the solutions independently of the output of the interaction and the submission should include a note denoting the nature of the collaboration. The use of generative AI tools to substantially complete sections of the assignments is not in line with the spirit of the assignments, and would be a violation of the [Honor Code](#).

```
[36]: # This mounts your Google Drive to the Colab VM.
from google.colab import drive
drive.mount('/content/drive')

# TODO: Enter the foldername in your Drive where you have saved the unzipped
# assignment folder, e.g. 'cs231n/assignments/assignment1/'
FOLDERNAME = 'cs231n/assignments/assignment1/'
assert FOLDERNAME is not None, "[!] Enter the foldername."

# Now that we've mounted your Drive, this ensures that
# the Python interpreter of the Colab VM can load
# python files from within it.
import sys
sys.path.append('/content/drive/My Drive/{}'.format(FOLDERNAME))

# This downloads the CIFAR-10 dataset to your Drive
# if it doesn't already exist.
%cd /content/drive/My\ Drive/$FOLDERNAME/cs231n/datasets/
!bash get_datasets.sh
%cd /content/drive/My\ Drive/$FOLDERNAME
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call `drive.mount("/content/drive", force_remount=True)`.

```
/content/drive/My Drive/cs231n/assignments/assignment1/cs231n/datasets
/content/drive/My Drive/cs231n/assignments/assignment1
```

1 k-Nearest Neighbor (kNN) exercise

Complete and hand in this completed worksheet (including its outputs and any supporting code outside of the worksheet) with your assignment submission. For more details see the [assignments page](#) on the course website.

The kNN classifier consists of two stages:

- During training, the classifier takes the training data and simply remembers it
- During testing, kNN classifies every test image by comparing to all training images and transferring the labels of the k most similar training examples
- The value of k is cross-validated

In this exercise you will implement these steps and understand the basic Image Classification pipeline, cross-validation, and gain proficiency in writing efficient, vectorized code.

```
[37]: # Run some setup code for this notebook.

import random
import numpy as np
from cs231n.data_utils import load_CIFAR10
import matplotlib.pyplot as plt

# This is a bit of magic to make matplotlib figures appear inline in the
↪notebook
# rather than in a new window.
%matplotlib inline
plt.rcParams['figure.figsize'] = (10.0, 8.0) # set default size of plots
plt.rcParams['image.interpolation'] = 'nearest'
plt.rcParams['image.cmap'] = 'gray'

# Some more magic so that the notebook will reload external python modules;
# see http://stackoverflow.com/questions/1907993/
↪autoreload-of-modules-in-ipython
%load_ext autoreload
%autoreload 2
```

The autoreload extension is already loaded. To reload it, use:

```
%reload_ext autoreload
```

```
[38]: # Load the raw CIFAR-10 data.
cifar10_dir = 'cs231n/datasets/cifar-10-batches-py'

# Cleaning up variables to prevent loading data multiple times (which may cause
↪memory issue)
try:
    del X_train, y_train
    del X_test, y_test
    print('Clear previously loaded data.')
except:
    pass

X_train, y_train, X_test, y_test = load_CIFAR10(cifar10_dir)
```

```
# As a sanity check, we print out the size of the training and test data.
print('Training data shape: ', X_train.shape)
print('Training labels shape: ', y_train.shape)
print('Test data shape: ', X_test.shape)
print('Test labels shape: ', y_test.shape)
```

Clear previously loaded data.

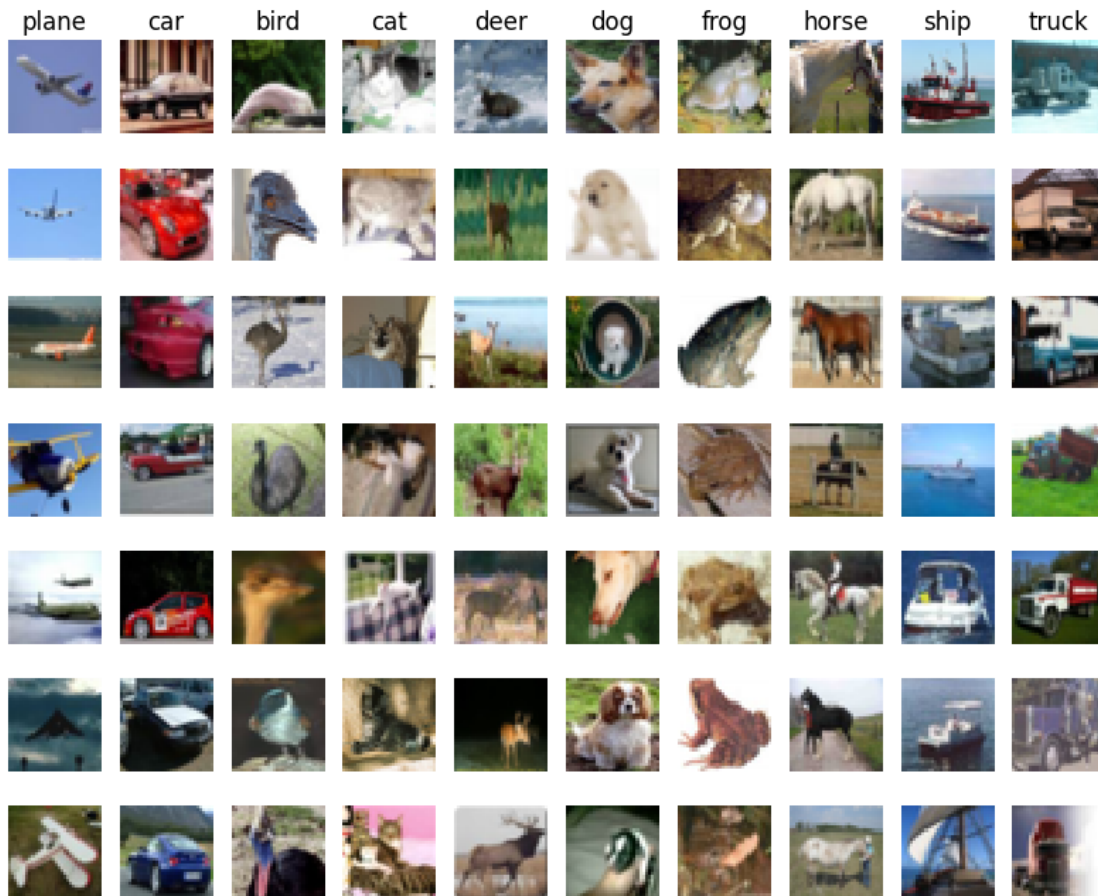
Training data shape: (50000, 32, 32, 3)

Training labels shape: (50000,)

Test data shape: (10000, 32, 32, 3)

Test labels shape: (10000,)

```
[39]: # Visualize some examples from the dataset.
# We show a few examples of training images from each class.
classes = ['plane', 'car', 'bird', 'cat', 'deer', 'dog', 'frog', 'horse', 'ship', 'truck']
num_classes = len(classes)
samples_per_class = 7
for y, cls in enumerate(classes):
    idxs = np.flatnonzero(y_train == y)
    idxs = np.random.choice(idxs, samples_per_class, replace=False)
    for i, idx in enumerate(idxs):
        plt_idx = i * num_classes + y + 1
        plt.subplot(samples_per_class, num_classes, plt_idx)
        plt.imshow(X_train[idx].astype('uint8'))
        plt.axis('off')
        if i == 0:
            plt.title(cls)
plt.show()
```



```
[40]: # Subsample the data for more efficient code execution in this exercise
num_training = 5000
mask = list(range(num_training))
X_train = X_train[mask]
y_train = y_train[mask]

num_test = 500
mask = list(range(num_test))
X_test = X_test[mask]
y_test = y_test[mask]

# Reshape the image data into rows
X_train = np.reshape(X_train, (X_train.shape[0], -1))
X_test = np.reshape(X_test, (X_test.shape[0], -1))
print(X_train.shape, X_test.shape)
```

(5000, 3072) (500, 3072)

```
[41]: from cs231n.classifiers import KNearestNeighbor

# Create a kNN classifier instance.
# Remember that training a kNN classifier is a noop:
# the Classifier simply remembers the data and does no further processing
classifier = KNearestNeighbor()
classifier.train(X_train, y_train)
```

We would now like to classify the test data with the kNN classifier. Recall that we can break down this process into two steps:

1. First we must compute the distances between all test examples and all train examples.
2. Given these distances, for each test example we find the k nearest examples and have them vote for the label

Lets begin with computing the distance matrix between all training and test examples. For example, if there are N_{tr} training examples and N_{te} test examples, this stage should result in a $N_{te} \times N_{tr}$ matrix where each element (i,j) is the distance between the i-th test and j-th train example.

Note: For the three distance computations that we require you to implement in this notebook, you may not use the `np.linalg.norm()` function that numpy provides.

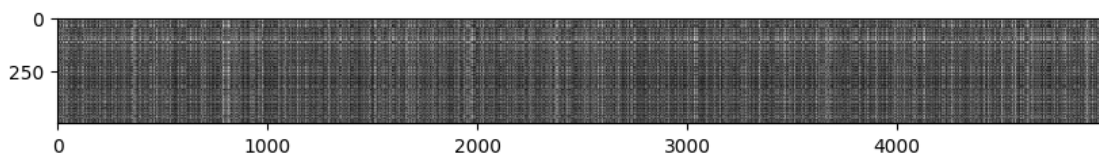
First, open `cs231n/classifiers/k_nearest_neighbor.py` and implement the function `compute_distances_two_loops` that uses a (very inefficient) double loop over all pairs of (test, train) examples and computes the distance matrix one element at a time.

```
[42]: # Open cs231n/classifiers/k_nearest_neighbor.py and implement
# compute_distances_two_loops.

# Test your implementation:
dists = classifier.compute_distances_two_loops(X_test)
print(dists.shape)
```

(500, 5000)

```
[43]: # We can visualize the distance matrix: each row is a single test example and
# its distances to training examples
plt.imshow(dists, interpolation='none')
plt.show()
```



Inline Question 1

Notice the structured patterns in the distance matrix, where some rows or columns are visibly brighter. (Note that with the default color scheme black indicates low distances while white indicates high distances.)

- What in the data is the cause behind the distinctly bright rows?
- What causes the columns?

Your Answer :

- Bright row is a result of higher L2 distance between a particular test image (represented by the row) with respect to majority of the train images. In other words, the broader background/color theme of the particular test image may not be matching most of the train images.
- Bright column is a result of higher L2 distance between a particular train image (represented by the column) with respect to majority of the test images. In other words, the broader background/color theme the particular train image may not be matching most of the test images.

```
[44]: # Now implement the function predict_labels and run the code below:
# We use k = 1 (which is Nearest Neighbor).
y_test_pred = classifier.predict_labels(dists, k=1)

# Compute and print the fraction of correctly predicted examples
num_correct = np.sum(y_test_pred == y_test)
accuracy = float(num_correct) / num_test
print('Got %d / %d correct => accuracy: %f' % (num_correct, num_test, accuracy))
```

Got 137 / 500 correct => accuracy: 0.274000

You should expect to see approximately 27% accuracy. Now lets try out a larger k, say k = 5:

```
[45]: y_test_pred = classifier.predict_labels(dists, k=5)
num_correct = np.sum(y_test_pred == y_test)
accuracy = float(num_correct) / num_test
print('Got %d / %d correct => accuracy: %f' % (num_correct, num_test, accuracy))
```

Got 139 / 500 correct => accuracy: 0.278000

You should expect to see a slightly better performance than with k = 1.

Inline Question 2

We can also use other distance metrics such as L1 distance. For pixel values $p_{ij}^{(k)}$ at location (i, j) of some image I_k ,

the mean μ across all pixels over all images is

$$\mu = \frac{1}{nhw} \sum_{k=1}^n \sum_{i=1}^h \sum_{j=1}^w p_{ij}^{(k)}$$

And the pixel-wise mean μ_{ij} across all images is

$$\mu_{ij} = \frac{1}{n} \sum_{k=1}^n p_{ij}^{(k)}.$$

The general standard deviation σ and pixel-wise standard deviation σ_{ij} is defined similarly.

Which of the following preprocessing steps will not change the performance of a Nearest Neighbor classifier that uses L1 distance? Select all that apply. To clarify, both training and test examples are preprocessed in the same way.

1. Subtracting the mean μ ($\tilde{p}_{ij}^{(k)} = p_{ij}^{(k)} - \mu$.)
2. Subtracting the per pixel mean μ_{ij} ($\tilde{p}_{ij}^{(k)} = p_{ij}^{(k)} - \mu_{ij}$.)
3. Subtracting the mean μ and dividing by the standard deviation σ .
4. Subtracting the pixel-wise mean μ_{ij} and dividing by the pixel-wise standard deviation σ_{ij} .
5. Rotating the coordinate axes of the data, which means rotating all the images by the same angle. Empty regions in the image caused by rotation are padded with a same pixel value and no interpolation is performed.

Your Answer : 1, 2, 3

Your Explanation :

Answer: # 1. Involves subtracting the same constant() from |a - b| L1 term, both cancel out thereby creating no effect on the performance of the nearest neighbour. 2. Has no effect on performance as μ_{ij} is still a constant among all the n images as far as pixels (i,j) is concerned thereby cancelling out in the L1 term. 3. Has no effect on performance as σ cancels out and dividing by σ scales down all the L1 distances equally thereby preserving the original nearest neighbours as is. 4. Changes the performance of the nearest neighbour classifier as L1 distance | (k1) - (k2) | component of every pixel (i,j) gets scaled by it's own $1/\sigma_{ij}$ value thereby suppressing or amplifying some of the pixels leading to a different overall distance affecting the nearest neighbor order. 5. Even though all the images are rotated by the same angle, L1 distance between 2 pixels, which measures a Manhattan distance (grid pattern) might change as the calculation is sensitive to rotation thereby potentially affecting the output of the nearest neighbor classifier.

[46]: *# Now lets speed up distance matrix computation by using partial vectorization
with one loop. Implement the function compute_distances_one_loop and run the
code below:*

```
dists_one = classifier.compute_distances_one_loop(X_test)
```

```
# To ensure that our vectorized implementation is correct, we make sure that it  
# agrees with the naive implementation. There are many ways to decide whether  
# two matrices are similar; one of the simplest is the Frobenius norm. In case  
# you haven't seen it before, the Frobenius norm of two matrices is the square  
# root of the squared sum of differences of all elements; in other words,  $\sqrt{\sum (a_{ij} - b_{ij})^2}$   
↪ reshape
```

```
# the matrices into vectors and compute the Euclidean distance between them.
```

```
difference = np.linalg.norm(dists - dists_one, ord='fro')
```

```
print('One loop difference was: %f' % (difference, ))
```

```
if difference < 0.001:
```

```

    print('Good! The distance matrices are the same')
else:
    print('Uh-oh! The distance matrices are different')

```

One loop difference was: 0.000000
 Good! The distance matrices are the same

```

[47]: # Now implement the fully vectorized version inside compute_distances_no_loops
      # and run the code
      dists_two = classifier.compute_distances_no_loops(X_test)

      # check that the distance matrix agrees with the one we computed before:
      difference = np.linalg.norm(dists - dists_two, ord='fro')
      print('No loop difference was: %f' % (difference, ))
      if difference < 0.001:
          print('Good! The distance matrices are the same')
      else:
          print('Uh-oh! The distance matrices are different')

```

No loop difference was: 0.000000
 Good! The distance matrices are the same

```

[48]: # Let's compare how fast the implementations are
      def time_function(f, *args):
          """
          Call a function f with args and return the time (in seconds) that it took
          to execute.
          """
          import time
          tic = time.time()
          f(*args)
          toc = time.time()
          return toc - tic

      two_loop_time = time_function(classifier.compute_distances_two_loops, X_test)
      print('Two loop version took %f seconds' % two_loop_time)

      one_loop_time = time_function(classifier.compute_distances_one_loop, X_test)
      print('One loop version took %f seconds' % one_loop_time)

      no_loop_time = time_function(classifier.compute_distances_no_loops, X_test)
      print('No loop version took %f seconds' % no_loop_time)

      # You should see significantly faster performance with the fully vectorized
      implementation!

      # NOTE: depending on what machine you're using,

```



```
# you might not see a speedup when you go from two loops to one loop,  
# and might even see a slow-down.
```

Two loop version took 22.619278 seconds
One loop version took 24.738756 seconds
No loop version took 0.376924 seconds

1.0.1 Cross-validation

We have implemented the k-Nearest Neighbor classifier but we set the value $k = 5$ arbitrarily. We will now determine the best value of this hyperparameter with cross-validation.

```
[49]: num_folds = 5  
k_choices = [1, 3, 5, 8, 10, 12, 15, 20, 50, 100]  
  
X_train_folds = []  
y_train_folds = []  
#####  
# Done: #  
# Split up the training data into folds. After splitting, X_train_folds and #  
# y_train_folds should each be lists of length num_folds, where #  
# y_train_folds[i] is the label vector for the points in X_train_folds[i]. #  
# Hint: Look up the numpy array_split function. #  
#####  
X_train_folds = np.array_split(X_train, num_folds)  
y_train_folds = np.array_split(y_train, num_folds)  
#####  
  
# A dictionary holding the accuracies for different values of k that we find  
# when running cross-validation. After running cross-validation,  
# k_to_accuracies[k] should be a list of length num_folds giving the different  
# accuracy values that we found when using that value of k.  
k_to_accuracies = {}  
  
#####  
# Done: #  
# Perform k-fold cross validation to find the best value of k. For each #  
# possible value of k, run the k-nearest-neighbor algorithm num_folds times, #  
# where in each case you use all but one of the folds as training data and the #  
# last fold as a validation set. Store the accuracies for all fold and all #  
# values of k in the k_to_accuracies dictionary. #  
#####  
for k in k_choices:  
    k_to_accuracies[k] = []  
    for i in range(num_folds):  
        X_train_set = np.concatenate(X_train_folds[:i] + X_train_folds[i+1:])  
        y_train_set = np.concatenate(y_train_folds[:i] + y_train_folds[i+1:])  
        x_valid_set = X_train_folds[i]
```

```

y_valid_set = y_train_folds[i]
classifier = KNearestNeighbor()
classifier.train(X_train_set, y_train_set)
y_valid_pred = classifier.predict(x_valid_set, k)
accuracy = np.mean(y_valid_pred == y_valid_set)
k_to_accuracies[k].append(accuracy)
# Print out the computed accuracies
for k in sorted(k_to_accuracies):
    for accuracy in k_to_accuracies[k]:
        print('k = %d, accuracy = %f' % (k, accuracy))

```

```

k = 1, accuracy = 0.263000
k = 1, accuracy = 0.257000
k = 1, accuracy = 0.264000
k = 1, accuracy = 0.278000
k = 1, accuracy = 0.266000
k = 3, accuracy = 0.239000
k = 3, accuracy = 0.249000
k = 3, accuracy = 0.240000
k = 3, accuracy = 0.266000
k = 3, accuracy = 0.254000
k = 5, accuracy = 0.248000
k = 5, accuracy = 0.266000
k = 5, accuracy = 0.280000
k = 5, accuracy = 0.292000
k = 5, accuracy = 0.280000
k = 8, accuracy = 0.262000
k = 8, accuracy = 0.282000
k = 8, accuracy = 0.273000
k = 8, accuracy = 0.290000
k = 8, accuracy = 0.273000
k = 10, accuracy = 0.265000
k = 10, accuracy = 0.296000
k = 10, accuracy = 0.276000
k = 10, accuracy = 0.284000
k = 10, accuracy = 0.280000
k = 12, accuracy = 0.260000
k = 12, accuracy = 0.295000
k = 12, accuracy = 0.279000
k = 12, accuracy = 0.283000
k = 12, accuracy = 0.280000
k = 15, accuracy = 0.252000
k = 15, accuracy = 0.289000
k = 15, accuracy = 0.278000
k = 15, accuracy = 0.282000
k = 15, accuracy = 0.274000
k = 20, accuracy = 0.270000

```

```

k = 20, accuracy = 0.279000
k = 20, accuracy = 0.279000
k = 20, accuracy = 0.282000
k = 20, accuracy = 0.285000
k = 50, accuracy = 0.271000
k = 50, accuracy = 0.288000
k = 50, accuracy = 0.278000
k = 50, accuracy = 0.269000
k = 50, accuracy = 0.266000
k = 100, accuracy = 0.256000
k = 100, accuracy = 0.270000
k = 100, accuracy = 0.263000
k = 100, accuracy = 0.256000
k = 100, accuracy = 0.263000

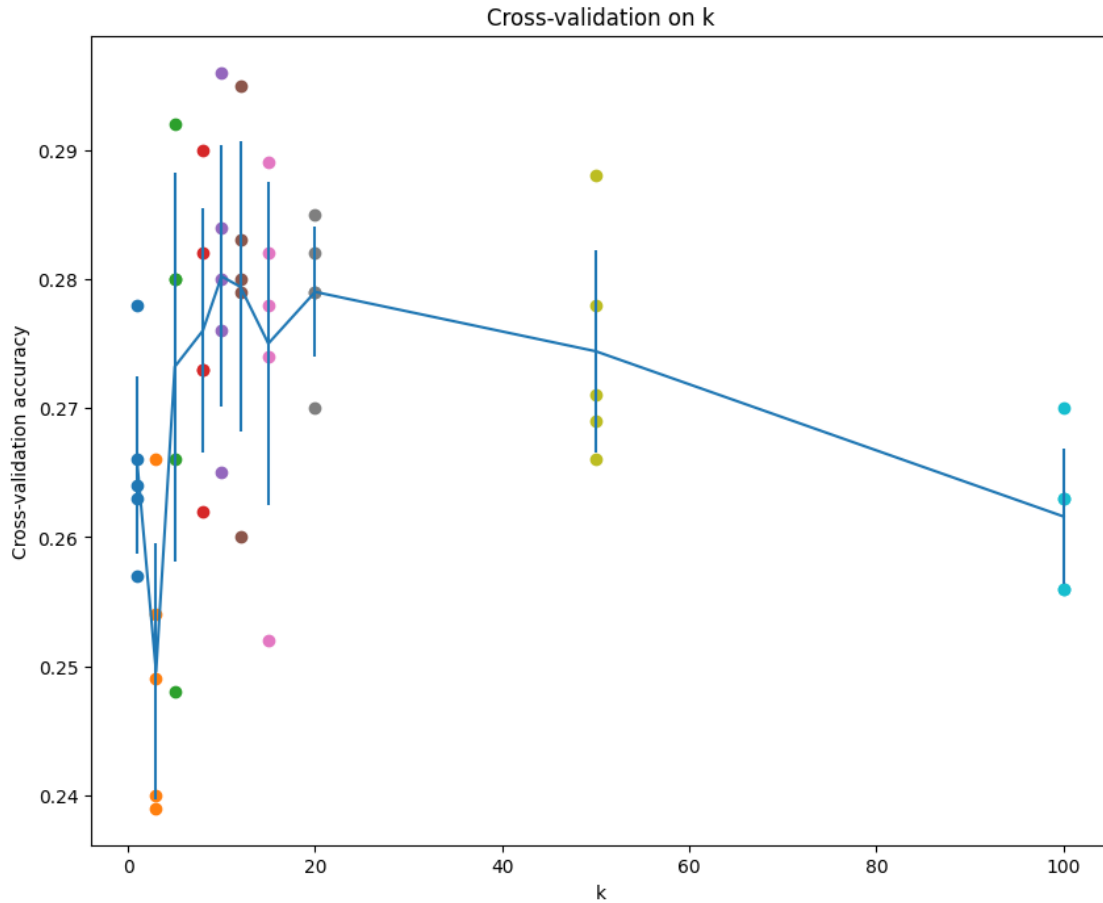
```

```

[50]: # plot the raw observations
      for k in k_choices:
          accuracies = k_to_accuracies[k]
          plt.scatter([k] * len(accuracies), accuracies)

      # plot the trend line with error bars that correspond to standard deviation
      accuracies_mean = np.array([np.mean(v) for k,v in sorted(k_to_accuracies.
          ↪items())])
      accuracies_std = np.array([np.std(v) for k,v in sorted(k_to_accuracies.
          ↪items())])
      plt.errorbar(k_choices, accuracies_mean, yerr=accuracies_std)
      plt.title('Cross-validation on k')
      plt.xlabel('k')
      plt.ylabel('Cross-validation accuracy')
      plt.show()

```



```
[51]: # Based on the cross-validation results above, choose the best value for k,
# retrain the classifier using all the training data, and test it on the test
# data. You should be able to get above 28% accuracy on the test data.
best_k = 10

classifier = KNearestNeighbor()
classifier.train(X_train, y_train)
y_test_pred = classifier.predict(X_test, k=best_k)

# Compute and display the accuracy
num_correct = np.sum(y_test_pred == y_test)
accuracy = float(num_correct) / num_test
print('Got %d / %d correct => accuracy: %f' % (num_correct, num_test, accuracy))
```

Got 141 / 500 correct => accuracy: 0.282000

Inline Question 3

Which of the following statements about k -Nearest Neighbor (k -NN) are true in a classification setting, and for all k ? Select all that apply. 1. The decision boundary of the k -NN classifier is

linear. 2. The training error of a 1-NN will always be lower than or equal to that of 5-NN. 3. The test error of a 1-NN will always be lower than that of a 5-NN. 4. The time needed to classify a test example with the k-NN classifier grows with the size of the training set. 5. None of the above.

Your Answer : 2, 4

Your Explanation : 1. False - KNN has a complex non-linear decision boundary that sort of follows the pixel values of the test and train data. 2. True - Training error of 1-NN is always zero as the nearest neighbor is itself. In case of 5-NN, it's possible that some, all or none of the neighbors vote for incorrect class leading to 0 or higher training error. Hence The training error of a 1-NN will always be lower than or equal to that of 5-NN holds true. 3. False - 1-NN overfits the training data (zero training error) and it has more test error than 5-NN usually, as higher N generalizes smoothly comparatively. Hence it's incorrect to claim that the test error of a 1-NN will always be lower than that of 5-NN. 4. True - As part of finding K-Nearest-Neighbors, KNN computes the distance of the test example with every sample in the training set. Hence, the time needed to classify a test sample grows as the size of the training set grows. 5. False

softmax

April 17, 2026

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```
[2]: # This mounts your Google Drive to the Colab VM.
from google.colab import drive
drive.mount('/content/drive')

# TODO: Enter the foldername in your Drive where you have saved the unzipped
# assignment folder, e.g. 'cs231n/assignments/assignment1/'
FOLDERNAME = 'cs231n/assignments/assignment1/'
assert FOLDERNAME is not None, "[!] Enter the foldername."

# Now that we've mounted your Drive, this ensures that
# the Python interpreter of the Colab VM can load
# python files from within it.
import sys
sys.path.append('/content/drive/My Drive/{}'.format(FOLDERNAME))

# This downloads the CIFAR-10 dataset to your Drive
# if it doesn't already exist.
%cd /content/drive/My\ Drive/$FOLDERNAME/cs231n/datasets/
!bash get_datasets.sh
%cd /content/drive/My\ Drive/$FOLDERNAME
```

```
Mounted at /content/drive
/content/drive/My Drive/cs231n/assignments/assignment1/cs231n/datasets
/content/drive/My Drive/cs231n/assignments/assignment1
```

1 Softmax Classifier exercise

Complete and hand in this completed worksheet (including its outputs and any supporting code outside of the worksheet) with your assignment submission. For more details see the [assignments page](#) on the course website.

In this exercise you will:

- implement a fully-vectorized **loss function** for the Softmax classifier.
- implement the fully-vectorized expression for its **analytic gradient**
- **check your implementation** using numerical gradient
- use a validation set to **tune the learning rate and regularization strength**
- **optimize** the loss function with **SGD**
- **visualize** the final learned weights

```
[3]: # Run some setup code for this notebook.
import random
import numpy as np
from cs231n.data_utils import load_CIFAR10
import matplotlib.pyplot as plt

# This is a bit of magic to make matplotlib figures appear inline in the
# notebook rather than in a new window.
%matplotlib inline
plt.rcParams['figure.figsize'] = (10.0, 8.0) # set default size of plots
plt.rcParams['image.interpolation'] = 'nearest'
plt.rcParams['image.cmap'] = 'gray'

# Some more magic so that the notebook will reload external python modules;
# see http://stackoverflow.com/questions/1907993/
↳ autoreload-of-modules-in-ipython
%load_ext autoreload
%autoreload 2
```

1.1 CIFAR-10 Data Loading and Preprocessing

```
[4]: # Load the raw CIFAR-10 data.
cifar10_dir = 'cs231n/datasets/cifar-10-batches-py'

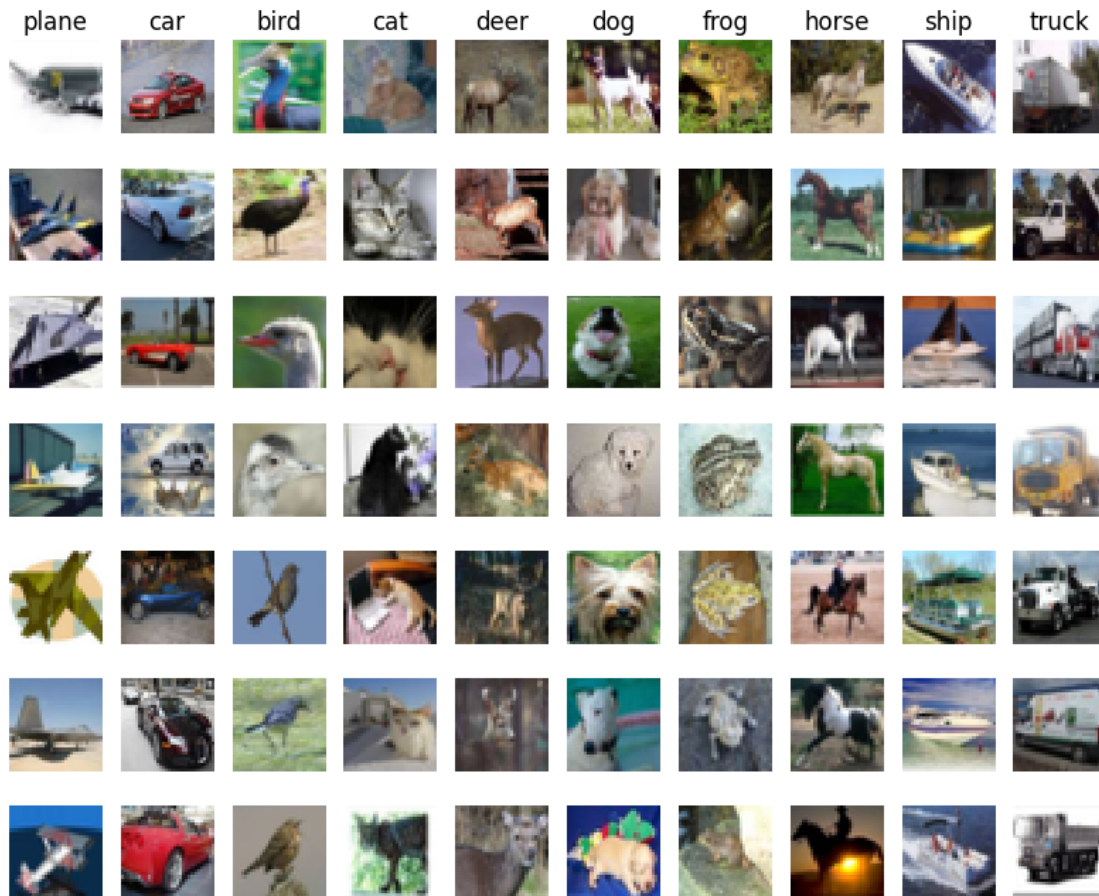
# Cleaning up variables to prevent loading data multiple times (which may cause
↳ memory issue)
try:
    del X_train, y_train
    del X_test, y_test
    print('Clear previously loaded data.')
except:
    pass

X_train, y_train, X_test, y_test = load_CIFAR10(cifar10_dir)

# As a sanity check, we print out the size of the training and test data.
print('Training data shape: ', X_train.shape)
print('Training labels shape: ', y_train.shape)
print('Test data shape: ', X_test.shape)
print('Test labels shape: ', y_test.shape)
```

```
Training data shape: (50000, 32, 32, 3)
Training labels shape: (50000,)
Test data shape: (10000, 32, 32, 3)
Test labels shape: (10000,)
```

```
[5]: # Visualize some examples from the dataset.
# We show a few examples of training images from each class.
classes = ['plane', 'car', 'bird', 'cat', 'deer', 'dog', 'frog', 'horse', '
↳ 'ship', 'truck']
num_classes = len(classes)
samples_per_class = 7
for y, cls in enumerate(classes):
    idxs = np.flatnonzero(y_train == y)
    idxs = np.random.choice(idxs, samples_per_class, replace=False)
    for i, idx in enumerate(idxs):
        plt_idx = i * num_classes + y + 1
        plt.subplot(samples_per_class, num_classes, plt_idx)
        plt.imshow(X_train[idx].astype('uint8'))
        plt.axis('off')
        if i == 0:
            plt.title(cls)
plt.show()
```

```
[6]: # Split the data into train, val, and test sets. In addition we will
# create a small development set as a subset of the training data;
# we can use this for development so our code runs faster.
num_training = 49000
num_validation = 1000
num_test = 1000
num_dev = 500

# Our validation set will be num_validation points from the original
# training set.
mask = range(num_training, num_training + num_validation)
X_val = X_train[mask]
y_val = y_train[mask]

# Our training set will be the first num_train points from the original
# training set.
mask = range(num_training)
X_train = X_train[mask]
y_train = y_train[mask]
```

```

# We will also make a development set, which is a small subset of
# the training set.
mask = np.random.choice(num_training, num_dev, replace=False)
X_dev = X_train[mask]
y_dev = y_train[mask]

# We use the first num_test points of the original test set as our
# test set.
mask = range(num_test)
X_test = X_test[mask]
y_test = y_test[mask]

print('Train data shape: ', X_train.shape)
print('Train labels shape: ', y_train.shape)
print('Validation data shape: ', X_val.shape)
print('Validation labels shape: ', y_val.shape)
print('Test data shape: ', X_test.shape)
print('Test labels shape: ', y_test.shape)

```

```

Train data shape: (49000, 32, 32, 3)
Train labels shape: (49000,)
Validation data shape: (1000, 32, 32, 3)
Validation labels shape: (1000,)
Test data shape: (1000, 32, 32, 3)
Test labels shape: (1000,)

```

```

[7]: # Preprocessing: reshape the image data into rows
X_train = np.reshape(X_train, (X_train.shape[0], -1))
X_val = np.reshape(X_val, (X_val.shape[0], -1))
X_test = np.reshape(X_test, (X_test.shape[0], -1))
X_dev = np.reshape(X_dev, (X_dev.shape[0], -1))

# As a sanity check, print out the shapes of the data
print('Training data shape: ', X_train.shape)
print('Validation data shape: ', X_val.shape)
print('Test data shape: ', X_test.shape)
print('dev data shape: ', X_dev.shape)

```

```

Training data shape: (49000, 3072)
Validation data shape: (1000, 3072)
Test data shape: (1000, 3072)
dev data shape: (500, 3072)

```

```

[8]: # Preprocessing: subtract the mean image
# first: compute the image mean based on the training data
mean_image = np.mean(X_train, axis=0)

```

```

print(mean_image[:10]) # print a few of the elements
plt.figure(figsize=(4,4))
plt.imshow(mean_image.reshape((32,32,3)).astype('uint8')) # visualize the mean_
    ↳image
plt.show()

# second: subtract the mean image from train and test data
X_train -= mean_image
X_val -= mean_image
X_test -= mean_image
X_dev -= mean_image

# third: append the bias dimension of ones (i.e. bias trick) so that our_
    ↳classifier
# only has to worry about optimizing a single weight matrix W.
X_train = np.hstack([X_train, np.ones((X_train.shape[0], 1))])
X_val = np.hstack([X_val, np.ones((X_val.shape[0], 1))])
X_test = np.hstack([X_test, np.ones((X_test.shape[0], 1))])
X_dev = np.hstack([X_dev, np.ones((X_dev.shape[0], 1))])

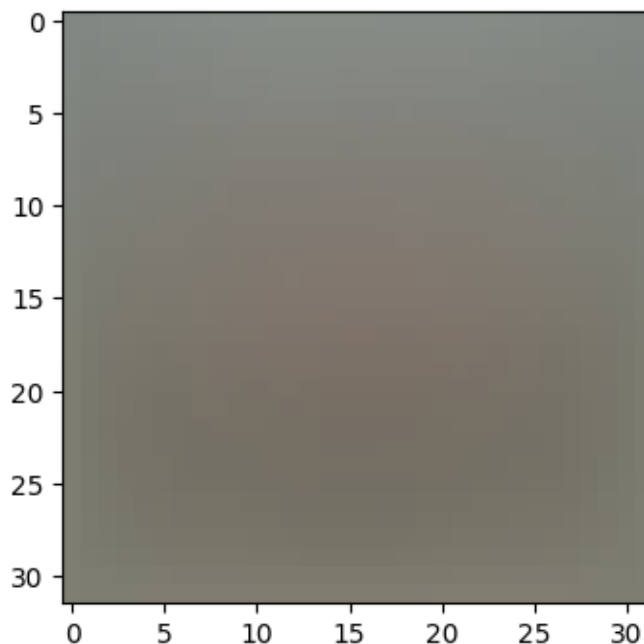
print(X_train.shape, X_val.shape, X_test.shape, X_dev.shape)

```

```

[130.64189796 135.98173469 132.47391837 130.05569388 135.34804082
 131.75402041 130.96055102 136.14328571 132.47636735 131.48467347]

```



```

(49000, 3073) (1000, 3073) (1000, 3073) (500, 3073)

```

1.2 Softmax Classifier

Your code for this section will all be written inside `cs231n/classifiers/softmax.py`.

As you can see, we have prefilled the function `softmax_loss_naive` which uses for loops to evaluate the softmax loss function.

```
[9]: # Evaluate the naive implementation of the loss we provided for you:
from cs231n.classifiers.softmax import softmax_loss_naive
import time

# generate a random Softmax classifier weight matrix of small numbers
W = np.random.randn(3073, 10) * 0.0001

loss, grad = softmax_loss_naive(W, X_dev, y_dev, 0.000005)
print('loss: %f' % (loss, ))

# As a rough sanity check, our loss should be something close to -log(0.1).
print('loss: %f' % loss)
print('sanity check: %f' % (-np.log(0.1)))
```

```
loss: 2.312623
loss: 2.312623
sanity check: 2.302585
```

Inline Question 1

Why do we expect our loss to be close to $-\log(0.1)$? Explain briefly.**

Your Answer : With randomly initialized weights that are close to zero (0.0001), the scores are nearly identical across all 10 classes. When near-equal scores are fed to softmax, denominator will be a multiple of numerator by a factor of # no. of classes (i.e 10). Thereby making the probability of each class as 0.1. Since our loss function is $-\log(p)$, we expect the loss to be $-\log(0.1)$ which is 2.302585.

The `grad` returned from the function above is right now all zero. Derive and implement the gradient for the softmax loss function and implement it inline inside the function `softmax_loss_naive`. You will find it helpful to interleave your new code inside the existing function.

To check that you have correctly implemented the gradient, you can numerically estimate the gradient of the loss function and compare the numeric estimate to the gradient that you computed. We have provided code that does this for you:

```
[10]: # Once you've implemented the gradient, recompute it with the code below
# and gradient check it with the function we provided for you

# Compute the loss and its gradient at W.
loss, grad = softmax_loss_naive(W, X_dev, y_dev, 0.0)

# Numerically compute the gradient along several randomly chosen dimensions, and
```

```

# compare them with your analytically computed gradient. The numbers should
↳ match
# almost exactly along all dimensions.
from cs231n.gradient_check import grad_check_sparse
f = lambda w: softmax_loss_naive(w, X_dev, y_dev, 0.0)[0]
grad_numerical = grad_check_sparse(f, W, grad)

# do the gradient check once again with regularization turned on
# you didn't forget the regularization gradient did you?
loss, grad = softmax_loss_naive(W, X_dev, y_dev, 5e1)
f = lambda w: softmax_loss_naive(w, X_dev, y_dev, 5e1)[0]
grad_numerical = grad_check_sparse(f, W, grad)

```

```

numerical: -2.199386 analytic: -2.199386, relative error: 1.597635e-08
numerical: 1.150050 analytic: 1.150050, relative error: 1.057509e-07
numerical: 2.290957 analytic: 2.290957, relative error: 2.916645e-09
numerical: -4.322312 analytic: -4.322312, relative error: 1.467402e-08
numerical: -3.030747 analytic: -3.030747, relative error: 2.827190e-09
numerical: -0.936083 analytic: -0.936083, relative error: 1.168378e-08
numerical: 0.441152 analytic: 0.441152, relative error: 5.267828e-08
numerical: 1.435062 analytic: 1.435062, relative error: 4.465499e-08
numerical: 3.094828 analytic: 3.094827, relative error: 1.928485e-08
numerical: -1.238493 analytic: -1.238493, relative error: 1.632088e-08
numerical: 1.440699 analytic: 1.440699, relative error: 9.617135e-09
numerical: -1.847894 analytic: -1.847894, relative error: 6.033119e-09
numerical: -1.307611 analytic: -1.307611, relative error: 1.179391e-08
numerical: -1.705212 analytic: -1.705212, relative error: 5.030390e-09
numerical: 1.303635 analytic: 1.303635, relative error: 6.976625e-09
numerical: 2.319101 analytic: 2.319100, relative error: 2.440548e-08
numerical: -2.365472 analytic: -2.365472, relative error: 9.328419e-09
numerical: -0.625603 analytic: -0.625603, relative error: 5.977490e-08
numerical: -0.647429 analytic: -0.647429, relative error: 4.244621e-08
numerical: 1.119309 analytic: 1.119309, relative error: 5.838490e-08

```

Inline Question 2

Although gradcheck is reliable softmax loss, it is possible that for SVM loss, once in a while, a dimension in the gradcheck will not match exactly. What could such a discrepancy be caused by? Is it a reason for concern? What is a simple example in one dimension where a svm loss gradient check could fail? How would change the margin affect of the frequency of this happening?

Note that SVM loss for a sample (x_i, y_i) is defined as:

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + \Delta)$$

where j iterates over all classes except the correct class y_i and s_j denotes the classifier score for j^{th} class. Δ is a scalar margin. For more information, refer to ‘Multiclass Support Vector Machine loss’ on [this](#) page.

Hint: the SVM loss function is not strictly speaking differentiable.

Your Answer : The loss function that SVM uses has a different left derivative (0) and right derivative (1) at 0. GradCheck computes $f(w+h)$ and $f(w-h)$ functions. If $w+h$ and $w-h$ falls onto either side of this non-differentiable point, then the function evaluations go in different directions making it non-agreeable with the analytical gradient.

Therefore the discrepancy is caused by the non-differential nature of the function at certain point (Zero in our case) and $w+h$ and $w-h$ being either side of the non-differentiability.

For example, $s_j - s_{y_i} + \Delta = 0$, $f(W+h)$ leads to gradient of 1 and $f(W-h)$ yields a gradient of 0 thereby numerical gradient will be ~ 0.5 where as analytical gradient in the code might be matching 0 or 1.

In reality though, the true gradient is undefined at this zero point. It's not 0 or 1 values are wrong, they are simply subgradients.

This situation is not a concern because the probability of hitting such a non-differentiable point is practically ~ 0 in the continuous floating values of the weights. Even a zero initialization of weights won't cause this issue as delta is non-zero (usually 1).

Each term in the SVM loss has exactly one kink (non-differentiable point), located where: $s_j - s_{y_i} = -\Delta$.

- if $\Delta = 0$, then $s_j - s_{y_i} = 0$; very common early in training when weights are small and scores are nearly equal. Hence highest frequency
- if $\Delta = 1$, then $s_j - s_{y_i} = -1$; score differences rarely reach -1 early in training (which is when gradcheck is run typically). Hence low frequency
- if $\Delta = \text{large}$, then $s_j - s_{y_i} = -(\text{large number})$; score differences almost never reach this far from 0 with small random weights. Hence lowest frequency

Frequency: $\Delta=0 > \Delta=1 > \Delta=\text{large}$

```
[11]: # Next implement the function softmax_loss_vectorized; for now only compute the
      ↪ loss;
      # we will implement the gradient in a moment.
      tic = time.time()
      loss_naive, grad_naive = softmax_loss_naive(W, X_dev, y_dev, 0.000005)
      toc = time.time()
      print('Naive loss: %e computed in %fs' % (loss_naive, toc - tic))

      from cs231n.classifiers.softmax import softmax_loss_vectorized
      tic = time.time()
      loss_vectorized, _ = softmax_loss_vectorized(W, X_dev, y_dev, 0.000005)
      toc = time.time()
      print('Vectorized loss: %e computed in %fs' % (loss_vectorized, toc - tic))

      # The losses should match but your vectorized implementation should be much
      ↪ faster.
      print('difference: %f' % (loss_naive - loss_vectorized))
```

Naive loss: 2.312623e+00 computed in 0.079355s

Vectorized loss: 2.312623e+00 computed in 0.028459s
difference: -0.000000

```
[12]: # Complete the implementation of softmax_loss_vectorized, and compute the
      ↪ gradient
      # of the loss function in a vectorized way.

      # The naive implementation and the vectorized implementation should match, but
      # the vectorized version should still be much faster.
      tic = time.time()
      _, grad_naive = softmax_loss_naive(W, X_dev, y_dev, 0.000005)
      toc = time.time()
      print('Naive loss and gradient: computed in %fs' % (toc - tic))

      tic = time.time()
      _, grad_vectorized = softmax_loss_vectorized(W, X_dev, y_dev, 0.000005)
      toc = time.time()
      print('Vectorized loss and gradient: computed in %fs' % (toc - tic))

      # The loss is a single number, so it is easy to compare the values computed
      # by the two implementations. The gradient on the other hand is a matrix, so
      # we use the Frobenius norm to compare them.
      difference = np.linalg.norm(grad_naive - grad_vectorized, ord='fro')
      print('difference: %f' % difference)
```

Naive loss and gradient: computed in 0.081730s
Vectorized loss and gradient: computed in 0.015723s
difference: 0.000000

1.2.1 Stochastic Gradient Descent

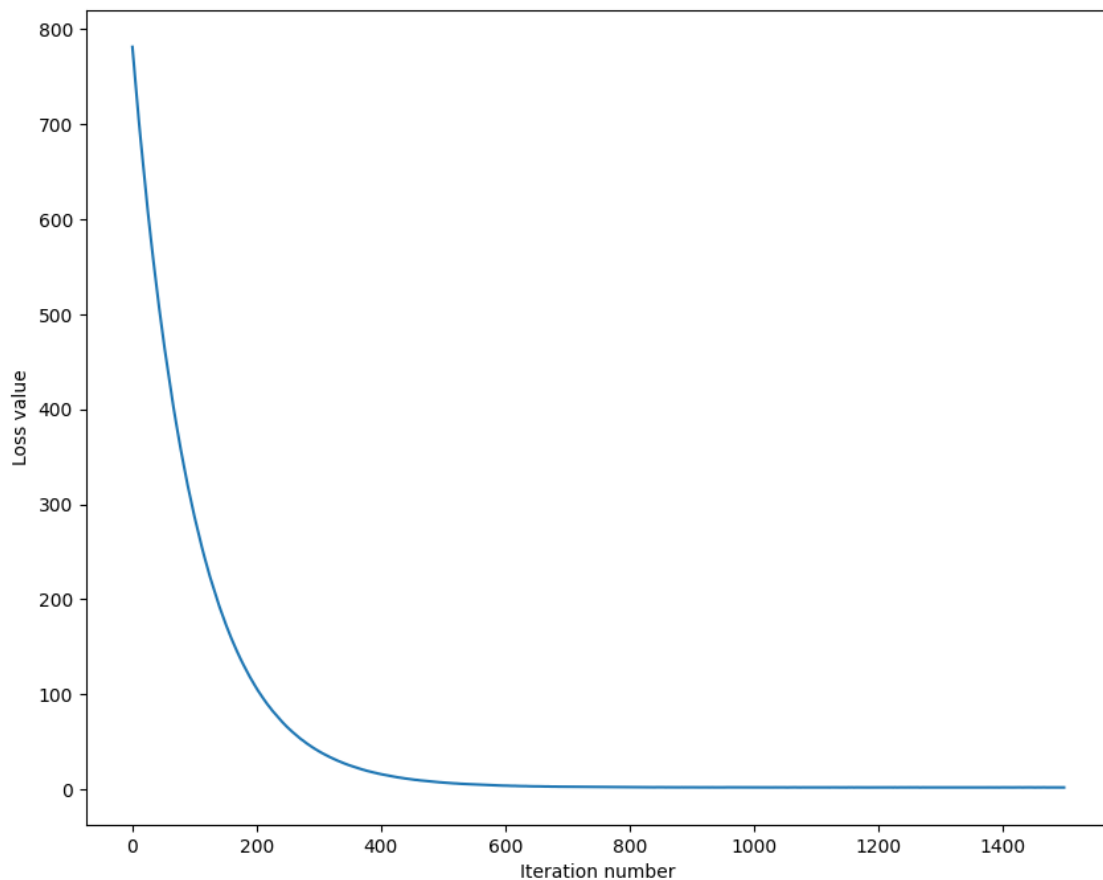
We now have vectorized and efficient expressions for the loss, the gradient and our gradient matches the numerical gradient. We are therefore ready to do SGD to minimize the loss. Your code for this part will be written inside `cs231n/classifiers/linear_classifier.py`.

```
[13]: # In the file linear_classifier.py, implement SGD in the function
      # LinearClassifier.train() and then run it with the code below.
      from cs231n.classifiers import Softmax
      softmax = Softmax()
      tic = time.time()
      loss_hist = softmax.train(X_train, y_train, learning_rate=1e-7, reg=2.5e4,
                               num_iters=1500, verbose=True)
      toc = time.time()
      print('That took %fs' % (toc - tic))
```

iteration 0 / 1500: loss 781.235241
iteration 100 / 1500: loss 286.689534
iteration 200 / 1500: loss 106.144281

```
iteration 300 / 1500: loss 40.212598
iteration 400 / 1500: loss 16.014315
iteration 500 / 1500: loss 7.189568
iteration 600 / 1500: loss 3.888081
iteration 700 / 1500: loss 2.805759
iteration 800 / 1500: loss 2.315483
iteration 900 / 1500: loss 2.161114
iteration 1000 / 1500: loss 2.105355
iteration 1100 / 1500: loss 2.098766
iteration 1200 / 1500: loss 2.063510
iteration 1300 / 1500: loss 2.096047
iteration 1400 / 1500: loss 2.108226
That took 13.092101s
```

```
[14]: # A useful debugging strategy is to plot the loss as a function of
      # iteration number:
      plt.plot(loss_hist)
      plt.xlabel('Iteration number')
      plt.ylabel('Loss value')
      plt.show()
```




```
[15]: # Write the LinearClassifier.predict function and evaluate the performance on
# both the training and validation set
# You should get validation accuracy of about 0.34 (> 0.33).
y_train_pred = softmax.predict(X_train)
print('training accuracy: %f' % (np.mean(y_train == y_train_pred), ))
y_val_pred = softmax.predict(X_val)
print('validation accuracy: %f' % (np.mean(y_val == y_val_pred), ))
```

```
training accuracy: 0.329061
validation accuracy: 0.345000
```

```
[16]: # Save the trained model for autograder.
softmax.save("softmax.npy")
```

softmax.npy saved.

```
[17]: # Use the validation set to tune hyperparameters (regularization strength and
# learning rate). You should experiment with different ranges for the learning
# rates and regularization strengths; if you are careful you should be able to
# get a classification accuracy of about 0.365 (> 0.36) on the validation set.

# Note: you may see runtime/overflow warnings during hyper-parameter search.
# This may be caused by extreme values, and is not a bug.

# results is dictionary mapping tuples of the form
# (learning_rate, regularization_strength) to tuples of the form
# (training_accuracy, validation_accuracy). The accuracy is simply the fraction
# of data points that are correctly classified.
results = {}
best_val = -1 # The highest validation accuracy that we have seen so far.
best_softmax = None # The Softmax object that achieved the highest validation
    ↪rate.

#####
# Done: #
# Write code that chooses the best hyperparameters by tuning on the validation #
# set. For each combination of hyperparameters, train a Softmax on the. #
# training set, compute its accuracy on the training and validation sets, and #
# store these numbers in the results dictionary. In addition, store the best #
# validation accuracy in best_val and the Softmax object that achieves this. #
# accuracy in best_softmax. #
# #
# Hint: You should use a small value for num_iters as you develop your #
# validation code so that the classifiers don't take much time to train; once #
# you are confident that your validation code works, you should rerun the #
# code with a larger value for num_iters. #
```

```
#####

# Provided as a reference. You may or may not want to change these
↳hyperparameters
learning_rates = [1e-7, 1e-6]
regularization_strengths = [2.5e4, 1e4]

for lr in learning_rates:
    for reg in regularization_strengths:
        # Train a softmax classifier
        softmax = Softmax()
        softmax.train(X_train, y_train, learning_rate=lr, reg=reg,
↳num_iters=2000)

        # Compute training accuracy
        y_train_pred = softmax.predict(X_train)
        train_accuracy = np.mean(y_train == y_train_pred)

        # Compute validation accuracy
        y_val_pred = softmax.predict(X_val)
        val_accuracy = np.mean(y_val == y_val_pred)

        # Store results
        results[(lr, reg)] = (train_accuracy, val_accuracy)

        # Update best
        if val_accuracy > best_val:
            best_val = val_accuracy
            best_softmax = softmax

# Print out results.
for lr, reg in sorted(results):
    train_accuracy, val_accuracy = results[(lr, reg)]
    print('lr %e reg %e train accuracy: %f val accuracy: %f' % (
        lr, reg, train_accuracy, val_accuracy))

print('best validation accuracy achieved during cross-validation: %f' %
↳best_val)
```

```
lr 1.000000e-07 reg 1.000000e+04 train accuracy: 0.357980 val accuracy: 0.374000
lr 1.000000e-07 reg 2.500000e+04 train accuracy: 0.331449 val accuracy: 0.351000
lr 1.000000e-06 reg 1.000000e+04 train accuracy: 0.345939 val accuracy: 0.358000
lr 1.000000e-06 reg 2.500000e+04 train accuracy: 0.319571 val accuracy: 0.331000
best validation accuracy achieved during cross-validation: 0.374000
```

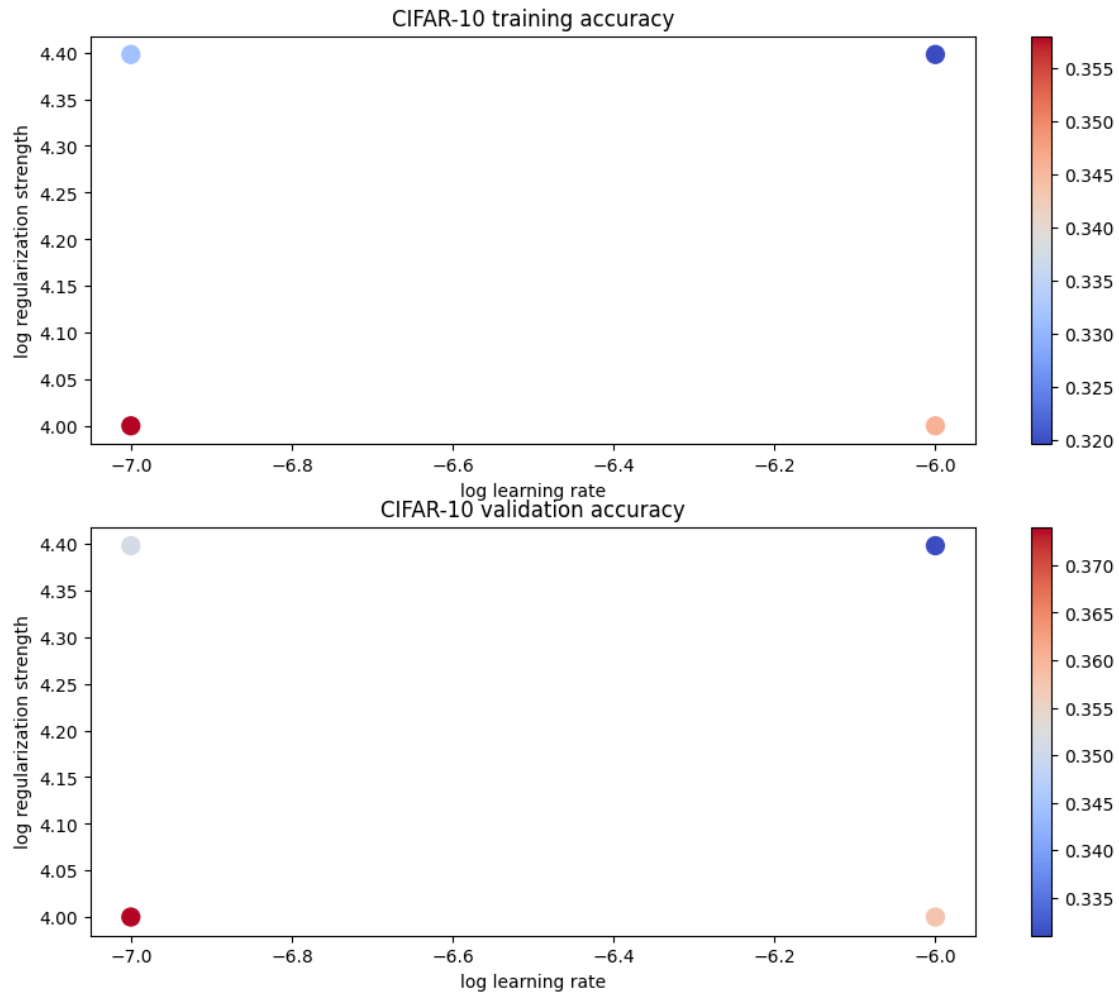
```
[18]: # Visualize the cross-validation results
import math
import pdb

# pdb.set_trace()

x_scatter = [math.log10(x[0]) for x in results]
y_scatter = [math.log10(x[1]) for x in results]

# plot training accuracy
marker_size = 100
colors = [results[x][0] for x in results]
plt.subplot(2, 1, 1)
plt.tight_layout(pad=3)
plt.scatter(x_scatter, y_scatter, marker_size, c=colors, cmap=plt.cm.coolwarm)
plt.colorbar()
plt.xlabel('log learning rate')
plt.ylabel('log regularization strength')
plt.title('CIFAR-10 training accuracy')

# plot validation accuracy
colors = [results[x][1] for x in results] # default size of markers is 20
plt.subplot(2, 1, 2)
plt.scatter(x_scatter, y_scatter, marker_size, c=colors, cmap=plt.cm.coolwarm)
plt.colorbar()
plt.xlabel('log learning rate')
plt.ylabel('log regularization strength')
plt.title('CIFAR-10 validation accuracy')
plt.show()
```



```
[19]: # Evaluate the best softmax on test set
y_test_pred = best_softmax.predict(X_test)
test_accuracy = np.mean(y_test == y_test_pred)
print('Softmax classifier on raw pixels final test set accuracy: %f' % test_accuracy)
```

Softmax classifier on raw pixels final test set accuracy: 0.365000

```
[20]: # Save best softmax model
best_softmax.save("best_softmax.npy")
```

best_softmax.npy saved.

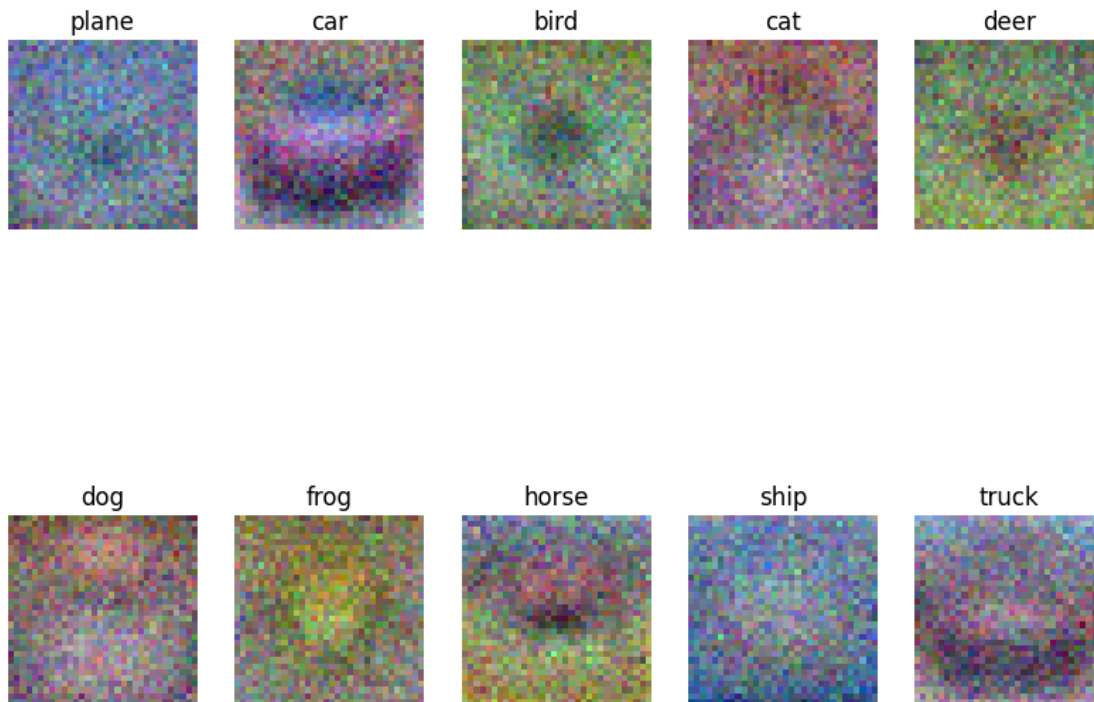
```
[21]: # Visualize the learned weights for each class.
# Depending on your choice of learning rate and regularization strength, these
may
```

```

# or may not be nice to look at.
w = best_softmax.W[:-1,:] # strip out the bias
w = w.reshape(32, 32, 3, 10)
w_min, w_max = np.min(w), np.max(w)
classes = ['plane', 'car', 'bird', 'cat', 'deer', 'dog', 'frog', 'horse', 'ship', 'truck']
for i in range(10):
    plt.subplot(2, 5, i + 1)

    # Rescale the weights to be between 0 and 255
    wimg = 255.0 * (w[:, :, :, i].squeeze() - w_min) / (w_max - w_min)
    plt.imshow(wimg.astype('uint8'))
    plt.axis('off')
    plt.title(classes[i])

```



Inline question 3

Describe what your visualized Softmax classifier weights look like, and offer a brief explanation for why they look the way they do.

Your Answer : The weight images are blurry, noisy, and pixelated. In spite of being far from the clear images, they loosely represent or capture the underlying class. This is because the learned weights for each class serve as the average template of the class. This average affect causes blurriness while hazily highlighting the underlying pattern of a particular class.

Inline Question 4 - True or False

Suppose the overall training loss is defined as the sum of the per-datapoint loss over all training examples. It is possible to add a new datapoint to a training set that would change the softmax loss, but leave the SVM loss unchanged.

Your Answer : True

Your Explanation : When per-datapoint loss is summed over all training examples, adding a new datapoint would change the softmax loss as softmax aims to represent the correct class probabilities closer to 1. In the case of SVM, if the correct class's score is ahead of all incorrect classes scores by at least Δ , and the new datapoint doesn't violate the margin, then SVM loss is unchanged.

Hence the answer is true!

[21] :

two_layer_net

April 17, 2026

Generative AI Use: For the purposes of the assignments, the use of generative AI is subject to the same policies regarding collaboration. Just as with other collaborators, each student must write down the solutions independently of the output of the interaction and the submission should include a note denoting the nature of the collaboration. The use of generative AI tools to substantially complete sections of the assignments is not in line with the spirit of the assignments, and would be a violation of the [Honor Code](#).

```
[28]: # This mounts your Google Drive to the Colab VM.
from google.colab import drive
drive.mount('/content/drive')

# TODO: Enter the foldername in your Drive where you have saved the unzipped
# assignment folder, e.g. 'cs231n/assignments/assignment1/'
FOLDERNAME = 'cs231n/assignments/assignment1/'
assert FOLDERNAME is not None, "[!] Enter the foldername."

# Now that we've mounted your Drive, this ensures that
# the Python interpreter of the Colab VM can load
# python files from within it.
import sys
sys.path.append('/content/drive/My Drive/{}'.format(FOLDERNAME))

# This downloads the CIFAR-10 dataset to your Drive
# if it doesn't already exist.
%cd /content/drive/My\ Drive/$FOLDERNAME/cs231n/datasets/
!bash get_datasets.sh
%cd /content/drive/My\ Drive/$FOLDERNAME
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call `drive.mount("/content/drive", force_remount=True)`.

```
/content/drive/My Drive/cs231n/assignments/assignment1/cs231n/datasets
/content/drive/My Drive/cs231n/assignments/assignment1
```

1 Fully-Connected Neural Nets

In this exercise we will implement fully-connected networks using a modular approach. For each layer we will implement a `forward` and a `backward` function. The `forward` function will receive inputs, weights, and other parameters and will return both an output and a `cache` object storing

data needed for the backward pass, like this:

```
def layer_forward(x, w):
    """ Receive inputs x and weights w """
    # Do some computations ...
    z = # ... some intermediate value
    # Do some more computations ...
    out = # the output

    cache = (x, w, z, out) # Values we need to compute gradients

    return out, cache
```

The backward pass will receive upstream derivatives and the `cache` object, and will return gradients with respect to the inputs and weights, like this:

```
def layer_backward(dout, cache):
    """
    Receive dout (derivative of loss with respect to outputs) and cache,
    and compute derivative with respect to inputs.
    """
    # Unpack cache values
    x, w, z, out = cache

    # Use values in cache to compute derivatives
    dx = # Derivative of loss with respect to x
    dw = # Derivative of loss with respect to w

    return dx, dw
```

After implementing a bunch of layers this way, we will be able to easily combine them to build classifiers with different architectures.

```
[ ]: # As usual, a bit of setup
from __future__ import print_function
import time
import numpy as np
import matplotlib.pyplot as plt
from cs231n.classifiers.fc_net import *
from cs231n.data_utils import get_CIFAR10_data
from cs231n.gradient_check import eval_numerical_gradient, \
    eval_numerical_gradient_array
from cs231n.solver import Solver

%matplotlib inline
plt.rcParams['figure.figsize'] = (10.0, 8.0) # set default size of plots
plt.rcParams['image.interpolation'] = 'nearest'
plt.rcParams['image.cmap'] = 'gray'
```



```
# for auto-reloading external modules
# see http://stackoverflow.com/questions/1907993/
# ↪ autoreload-of-modules-in-ipython
%load_ext autoreload
%autoreload 2

def rel_error(x, y):
    """ returns relative error """
    return np.max(np.abs(x - y) / (np.maximum(1e-8, np.abs(x) + np.abs(y))))
```

```
[ ]: # Load the (preprocessed) CIFAR10 data.
```

```
data = get_CIFAR10_data()
for k, v in list(data.items()):
    print('%s: ' % k, v.shape)
```

```
('X_train: ', (49000, 3, 32, 32))
('y_train: ', (49000,))
('X_val: ', (1000, 3, 32, 32))
('y_val: ', (1000,))
('X_test: ', (1000, 3, 32, 32))
('y_test: ', (1000,))
```

2 Affine layer: forward

Open the file `cs231n/layers.py` and implement the `affine_forward` function.

Once you are done you can test your implementation by running the following:

```
[ ]: # Test the affine_forward function
```

```
num_inputs = 2
input_shape = (4, 5, 6)
output_dim = 3

input_size = num_inputs * np.prod(input_shape)
weight_size = output_dim * np.prod(input_shape)

x = np.linspace(-0.1, 0.5, num=input_size).reshape(num_inputs, *input_shape)
w = np.linspace(-0.2, 0.3, num=weight_size).reshape(np.prod(input_shape),
    ↪ output_dim)
b = np.linspace(-0.3, 0.1, num=output_dim)

out, _ = affine_forward(x, w, b)
correct_out = np.array([[ 1.49834967,  1.70660132,  1.91485297],
    [ 3.25553199,  3.5141327,  3.77273342]])
```

```
# Compare your output with ours. The error should be around e-9 or less.
print('Testing affine_forward function:')
print('difference: ', rel_error(out, correct_out))
```

```
Testing affine_forward function:
difference: 9.769849468192957e-10
```

3 Affine layer: backward

Now implement the `affine_backward` function and test your implementation using numeric gradient checking.

```
[ ]: # Test the affine_backward function
np.random.seed(231)
x = np.random.randn(10, 2, 3)
w = np.random.randn(6, 5)
b = np.random.randn(5)
dout = np.random.randn(10, 5)

dx_num = eval_numerical_gradient_array(lambda x: affine_forward(x, w, b)[0], x,
    ↪dout)
dw_num = eval_numerical_gradient_array(lambda w: affine_forward(x, w, b)[0], w,
    ↪dout)
db_num = eval_numerical_gradient_array(lambda b: affine_forward(x, w, b)[0], b,
    ↪dout)

_, cache = affine_forward(x, w, b)
dx, dw, db = affine_backward(dout, cache)

# The error should be around e-10 or less
print('Testing affine_backward function:')
print('dx error: ', rel_error(dx_num, dx))
print('dw error: ', rel_error(dw_num, dw))
print('db error: ', rel_error(db_num, db))
```

```
Testing affine_backward function:
dx error: 5.399100368651805e-11
dw error: 9.904211865398145e-11
db error: 2.4122867568119087e-11
```

4 ReLU activation: forward

Implement the forward pass for the ReLU activation function in the `relu_forward` function and test your implementation using the following:

```
[ ]: # Test the relu_forward function
```

```

x = np.linspace(-0.5, 0.5, num=12).reshape(3, 4)

out, _ = relu_forward(x)
correct_out = np.array([[ 0.,          0.,          0.,          0.,          ],
                        [ 0.,          0.,          0.04545455, 0.13636364,],
                        [ 0.22727273, 0.31818182, 0.40909091, 0.5,          ]])

# Compare your output with ours. The error should be on the order of e-8
print('Testing relu_forward function:')
print('difference: ', rel_error(out, correct_out))

```

```

Testing relu_forward function:
difference:  4.999999798022158e-08

```

5 ReLU activation: backward

Now implement the backward pass for the ReLU activation function in the `relu_backward` function and test your implementation using numeric gradient checking:

```

[ ]: np.random.seed(231)
x = np.random.randn(10, 10)
dout = np.random.randn(*x.shape)

dx_num = eval_numerical_gradient_array(lambda x: relu_forward(x)[0], x, dout)

_, cache = relu_forward(x)
dx = relu_backward(dout, cache)

# The error should be on the order of e-12
print('Testing relu_backward function:')
print('dx error: ', rel_error(dx_num, dx))

```

```

Testing relu_backward function:
dx error:  3.2756349136310288e-12

```

5.1 Inline Question 1:

We've only asked you to implement ReLU, but there are a number of different activation functions that one could use in neural networks, each with its pros and cons. In particular, an issue commonly seen with activation functions is getting zero (or close to zero) gradient flow during backpropagation. Which of the following activation functions have this problem? If you consider these functions in the one dimensional case, what types of input would lead to this behaviour? 1. Sigmoid 2. ReLU 3. Leaky ReLU

Your Answer : Both Sigmoid and ReLU

$$f'(x) = f(x)(1 - f(x))$$

- when x is very positive, $f'(x) = 0$

- when x is very negative, $\sigma'(x) = 0$

$\text{ReLU}'(x) = 0$ when $x \leq 0$

where $\sigma'(x)$ is derivative of $\sigma(x)$ and $\text{ReLU}'(x)$ is derivative of $\text{ReLU}(x)$

Leaky ReLU doesn't have this problem as leakyReLU returns x when $x \leq 0$ where α is a small constant, typically 0.01.

6 “Sandwich” layers

There are some common patterns of layers that are frequently used in neural nets. For example, affine layers are frequently followed by a ReLU nonlinearity. To make these common patterns easy, we define several convenience layers in the file `cs231n/layer_utils.py`.

For now take a look at the `affine_relu_forward` and `affine_relu_backward` functions, and run the following to numerically gradient check the backward pass:

```
[ ]: from cs231n.layer_utils import affine_relu_forward, affine_relu_backward
np.random.seed(231)
x = np.random.randn(2, 3, 4)
w = np.random.randn(12, 10)
b = np.random.randn(10)
dout = np.random.randn(2, 10)

out, cache = affine_relu_forward(x, w, b)
dx, dw, db = affine_relu_backward(dout, cache)

dx_num = eval_numerical_gradient_array(lambda x: affine_relu_forward(x, w, b),
                                       x, dout)
dw_num = eval_numerical_gradient_array(lambda w: affine_relu_forward(x, w, b),
                                       w, dout)
db_num = eval_numerical_gradient_array(lambda b: affine_relu_forward(x, w, b),
                                       b, dout)

# Relative error should be around e-10 or less
print('Testing affine_relu_forward and affine_relu_backward:')
print('dx error: ', rel_error(dx_num, dx))
print('dw error: ', rel_error(dw_num, dw))
print('db error: ', rel_error(db_num, db))
```

Testing affine_relu_forward and affine_relu_backward:

dx error: 2.299579177309368e-11

dw error: 8.162011105764925e-11

db error: 7.826724021458994e-12

7 Loss layers: Softmax

Now implement the loss and gradient for softmax in the `softmax_loss` function in `cs231n/layers.py`. These should be similar to what you implemented in `cs231n/classifiers/softmax.py`. Other loss functions (e.g. `svm_loss`) can also be implemented in a modular way, however, it is not required for this assignment.

You can make sure that the implementations are correct by running the following:

```
[ ]: np.random.seed(231)
num_classes, num_inputs = 10, 50
x = 0.001 * np.random.randn(num_inputs, num_classes)
y = np.random.randint(num_classes, size=num_inputs)

dx_num = eval_numerical_gradient(lambda x: softmax_loss(x, y)[0], x,
    verbose=False)
loss, dx = softmax_loss(x, y)

# Test softmax_loss function. Loss should be close to 2.3 and dx error should
    be around e-8
print('\nTesting softmax_loss:')
print('loss: ', loss)
print('dx error: ', rel_error(dx_num, dx))
```

```
Testing softmax_loss:
loss:  2.3025458445007376
dx error:  8.234144091578429e-09
```

8 Two-layer network

Open the file `cs231n/classifiers/fc_net.py` and complete the implementation of the `TwoLayerNet` class. Read through it to make sure you understand the API. You can run the cell below to test your implementation.

```
[ ]: np.random.seed(231)
N, D, H, C = 3, 5, 50, 7
X = np.random.randn(N, D)
y = np.random.randint(C, size=N)

std = 1e-3
model = TwoLayerNet(input_dim=D, hidden_dim=H, num_classes=C, weight_scale=std)

print('Testing initialization ... ')
W1_std = abs(model.params['W1'].std() - std)
b1 = model.params['b1']
W2_std = abs(model.params['W2'].std() - std)
```

```

b2 = model.params['b2']
assert W1_std < std / 10, 'First layer weights do not seem right'
assert np.all(b1 == 0), 'First layer biases do not seem right'
assert W2_std < std / 10, 'Second layer weights do not seem right'
assert np.all(b2 == 0), 'Second layer biases do not seem right'

print('Testing test-time forward pass ... ')
model.params['W1'] = np.linspace(-0.7, 0.3, num=D*H).reshape(D, H)
model.params['b1'] = np.linspace(-0.1, 0.9, num=H)
model.params['W2'] = np.linspace(-0.3, 0.4, num=H*C).reshape(H, C)
model.params['b2'] = np.linspace(-0.9, 0.1, num=C)
X = np.linspace(-5.5, 4.5, num=N*D).reshape(D, N).T
scores = model.loss(X)
correct_scores = np.asarray(
    [[11.53165108, 12.2917344, 13.05181771, 13.81190102, 14.57198434, 15.
    ↪33206765, 16.09215096],
    [12.05769098, 12.74614105, 13.43459113, 14.1230412, 14.81149128, 15.
    ↪49994135, 16.18839143],
    [12.58373087, 13.20054771, 13.81736455, 14.43418138, 15.05099822, 15.
    ↪66781506, 16.2846319 ]])
scores_diff = np.abs(scores - correct_scores).sum()
assert scores_diff < 1e-6, 'Problem with test-time forward pass'

print('Testing training loss (no regularization)')
y = np.asarray([0, 5, 1])
loss, grads = model.loss(X, y)
correct_loss = 3.4702243556
assert abs(loss - correct_loss) < 1e-10, 'Problem with training-time loss'

model.reg = 1.0
loss, grads = model.loss(X, y)
correct_loss = 26.5948426952
assert abs(loss - correct_loss) < 1e-10, 'Problem with regularization loss'

# Errors should be around e-7 or less
for reg in [0.0, 0.7]:
    print('Running numeric gradient check with reg = ', reg)
    model.reg = reg
    loss, grads = model.loss(X, y)

    for name in sorted(grads):
        f = lambda _: model.loss(X, y)[0]
        grad_num = eval_numerical_gradient(f, model.params[name], verbose=False)
        print('%s relative error: %.2e' % (name, rel_error(grad_num, grads[name])))

```

Testing initialization ...

Testing test-time forward pass ...

```

Testing training loss (no regularization)
Running numeric gradient check with reg = 0.0
W1 relative error: 1.83e-08
W2 relative error: 3.20e-10
b1 relative error: 9.83e-09
b2 relative error: 4.33e-10
Running numeric gradient check with reg = 0.7
W1 relative error: 2.53e-07
W2 relative error: 2.85e-08
b1 relative error: 1.56e-08
b2 relative error: 9.09e-10

```

9 Solver

Open the file `cs231n/solver.py` and read through it to familiarize yourself with the API. After doing so, use a `Solver` instance to train a `TwoLayerNet` that achieves about 36% accuracy on the validation set.

```

[ ]: input_size = 32 * 32 * 3
     hidden_size = 50
     num_classes = 10
     #####
     # Done: Use a Solver instance to train a TwoLayerNet that achieves about 36% #
     # accuracy on the validation set.                                     #
     #####
     #####
     #                               END OF YOUR CODE                               #
     #####

model = TwoLayerNet(input_size, hidden_size, num_classes, reg=1e-3)

solver = Solver(model, data,
    optim_config={
        'learning_rate': 1e-3,
    },
    lr_decay=0.95,
    num_epochs=10,
    batch_size=200,
    print_every=100,
)

solver.train()

```

```

(Iteration 1 / 2450) loss: 2.301634
(Epoch 0 / 10) train acc: 0.153000; val_acc: 0.146000
(Iteration 101 / 2450) loss: 1.826498
(Iteration 201 / 2450) loss: 1.654066

```

```
(Epoch 1 / 10) train acc: 0.405000; val_acc: 0.400000
(Iteration 301 / 2450) loss: 1.643514
(Iteration 401 / 2450) loss: 1.510135
(Epoch 2 / 10) train acc: 0.460000; val_acc: 0.456000
(Iteration 501 / 2450) loss: 1.544156
(Iteration 601 / 2450) loss: 1.446108
(Iteration 701 / 2450) loss: 1.629429
(Epoch 3 / 10) train acc: 0.477000; val_acc: 0.469000
(Iteration 801 / 2450) loss: 1.295441
(Iteration 901 / 2450) loss: 1.617212
(Epoch 4 / 10) train acc: 0.492000; val_acc: 0.488000
(Iteration 1001 / 2450) loss: 1.498749
(Iteration 1101 / 2450) loss: 1.393832
(Iteration 1201 / 2450) loss: 1.442740
(Epoch 5 / 10) train acc: 0.485000; val_acc: 0.494000
(Iteration 1301 / 2450) loss: 1.240158
(Iteration 1401 / 2450) loss: 1.485876
(Epoch 6 / 10) train acc: 0.519000; val_acc: 0.480000
(Iteration 1501 / 2450) loss: 1.445548
(Iteration 1601 / 2450) loss: 1.389010
(Iteration 1701 / 2450) loss: 1.381947
(Epoch 7 / 10) train acc: 0.520000; val_acc: 0.497000
(Iteration 1801 / 2450) loss: 1.338639
(Iteration 1901 / 2450) loss: 1.376646
(Epoch 8 / 10) train acc: 0.565000; val_acc: 0.488000
(Iteration 2001 / 2450) loss: 1.261308
(Iteration 2101 / 2450) loss: 1.433225
(Iteration 2201 / 2450) loss: 1.128798
(Epoch 9 / 10) train acc: 0.548000; val_acc: 0.490000
(Iteration 2301 / 2450) loss: 1.209354
(Iteration 2401 / 2450) loss: 1.342911
(Epoch 10 / 10) train acc: 0.530000; val_acc: 0.495000
```

10 Debug the training

With the default parameters we provided above, you should get a validation accuracy of about 0.36 on the validation set. This isn't very good.

One strategy for getting insight into what's wrong is to plot the loss function and the accuracies on the training and validation sets during optimization.

Another strategy is to visualize the weights that were learned in the first layer of the network. In most neural networks trained on visual data, the first layer weights typically show some visible structure when visualized.

```
[ ]: # Run this cell to visualize training loss and train / val accuracy

plt.subplot(2, 1, 1)
```

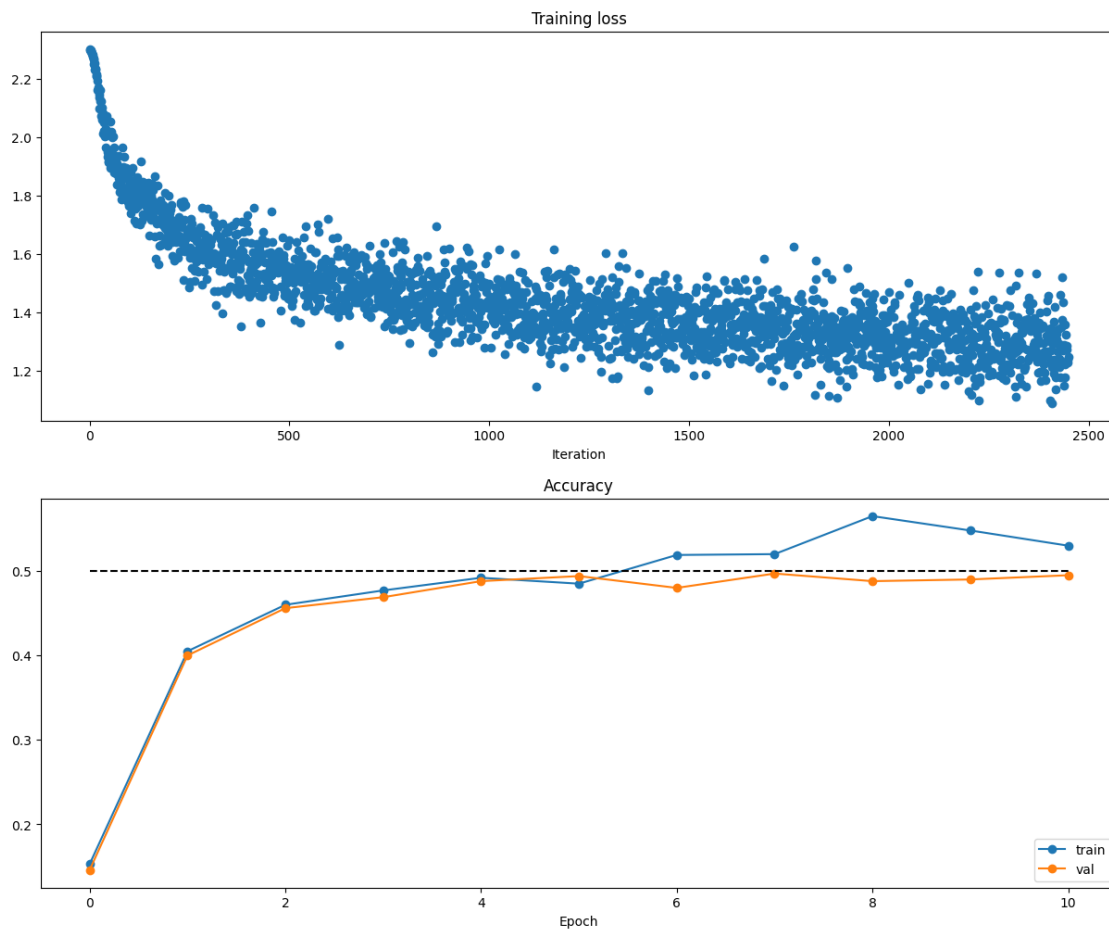


```

plt.title('Training loss')
plt.plot(solver.loss_history, 'o')
plt.xlabel('Iteration')

plt.subplot(2, 1, 2)
plt.title('Accuracy')
plt.plot(solver.train_acc_history, '-o', label='train')
plt.plot(solver.val_acc_history, '-o', label='val')
plt.plot([0.5] * len(solver.val_acc_history), 'k--')
plt.xlabel('Epoch')
plt.legend(loc='lower right')
plt.gcf().set_size_inches(15, 12)
plt.show()

```



```

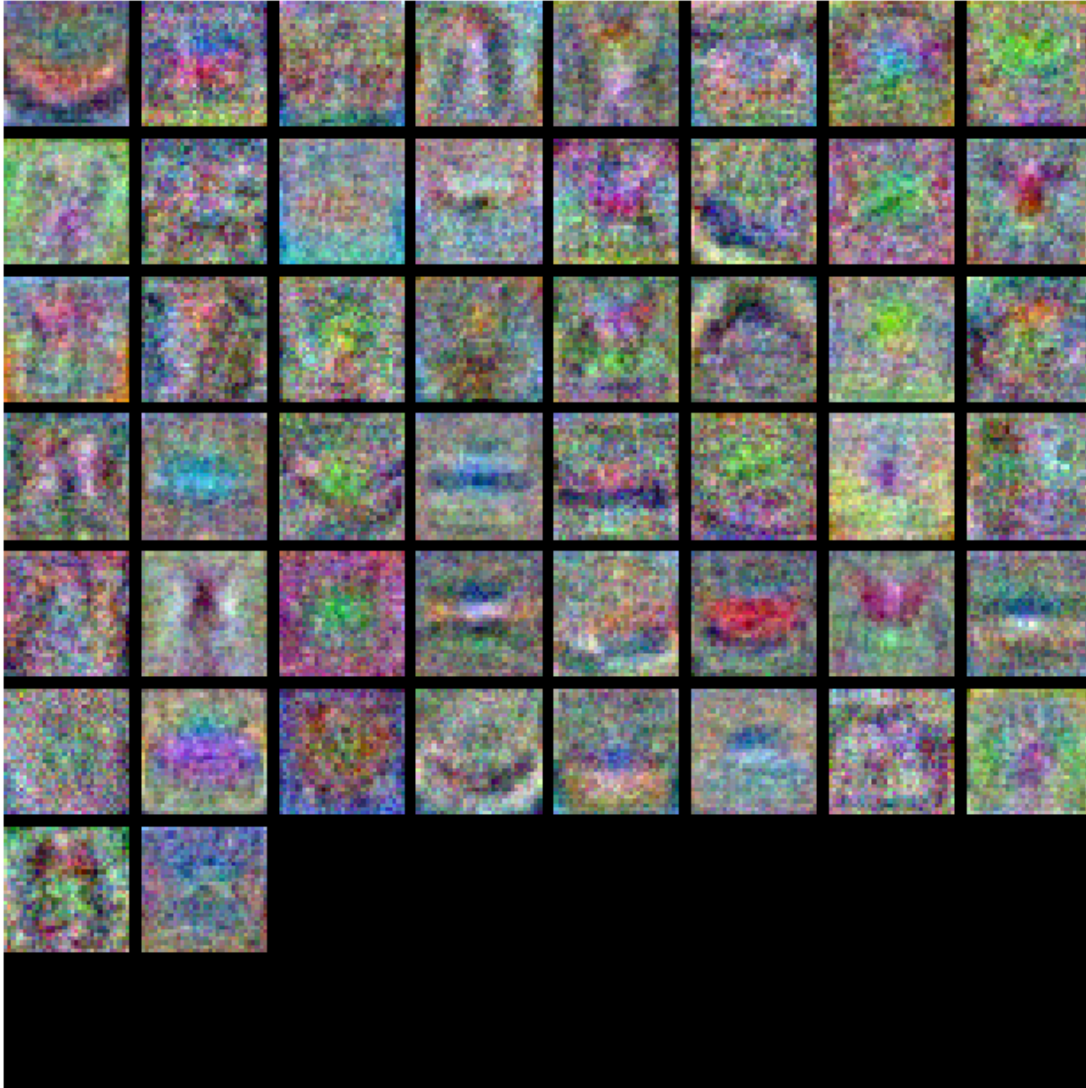
[ ]: from cs231n.vis_utils import visualize_grid

# Visualize the weights of the network

```

```
def show_net_weights(net):
    W1 = net.params['W1']
    W1 = W1.reshape(3, 32, 32, -1).transpose(3, 1, 2, 0)
    plt.imshow(visualize_grid(W1, padding=3).astype('uint8'))
    plt.gca().axis('off')
    plt.show()

show_net_weights(model)
```



11 Tune your hyperparameters

What's wrong?. Looking at the visualizations above, we see that the loss is decreasing more or less linearly, which seems to suggest that the learning rate may be too low. Moreover, there is no gap between the training and validation accuracy, suggesting that the model we used has low capacity, and that we should increase its size. On the other hand, with a very large model we would expect to see more overfitting, which would manifest itself as a very large gap between the training and validation accuracy.

Tuning. Tuning the hyperparameters and developing intuition for how they affect the final performance is a large part of using Neural Networks, so we want you to get a lot of practice. Below, you should experiment with different values of the various hyperparameters, including hidden layer size, learning rate, number of training epochs, and regularization strength. You might also consider tuning the learning rate decay, but you should be able to get good performance using the default value.

Approximate results. You should be aim to achieve a classification accuracy of greater than 48% on the validation set. Our best network gets over 52% on the validation set.

Experiment: Your goal in this exercise is to get as good of a result on CIFAR-10 as you can (52% could serve as a reference), with a fully-connected Neural Network. Feel free to implement your own techniques (e.g. PCA to reduce dimensionality, or adding dropout, or adding features to the solver, etc.).

```
[ ]: from sklearn.decomposition import PCA

best_model = None
best_val_acc = 0

# flatten once outside loop - doesn't change
X_train_flat = data['X_train'].reshape(data['X_train'].shape[0], -1)
X_val_flat    = data['X_val'].reshape(data['X_val'].shape[0], -1)
X_test_flat   = data['X_test'].reshape(data['X_test'].shape[0], -1)

learning_rates = [1e-3]
hidden_sizes = [300, 400, 500]
regs = [1e-4, 1e-5]
num_epochs = [20, 25]
pca_components = [50, 100, 200]

for n_comp in pca_components:
    pca = PCA(n_components=n_comp)          # uses n_comp
    X_train_pca = pca.fit_transform(X_train_flat)
    X_val_pca    = pca.transform(X_val_flat)
    X_test_pca   = pca.transform(X_test_flat)

    data_pca = {
        'X_train': X_train_pca, 'y_train': data['y_train'],
        'X_val':    X_val_pca,   'y_val':    data['y_val'],
```

```

        'X_test': X_test_pca, 'y_test': data['y_test'],
    }

    for lr in learning_rates:
        for hs in hidden_sizes:
            for reg in regs:
                for epochs in num_epochs:
                    model = TwoLayerNet(input_dim=n_comp, hidden_dim=hs,
                                         num_classes=num_classes, reg=reg)
                    solver = Solver(model, data_pca,
                                   optim_config={'learning_rate': lr},
                                   lr_decay=0.95,
                                   num_epochs=epochs,
                                   batch_size=200,
                                   verbose=False,
                                   )
                    solver.train()

                    val_acc = solver.best_val_acc
                    print(f'pca={n_comp}, lr={lr}, hs={hs}, reg={reg}, val_
↪ epochs={epochs} → val_acc={val_acc:.3f}')

                    if val_acc > best_val_acc:
                        best_val_acc = val_acc
                        best_model = model
                        best_pca = pca

print(f'\nbest val accuracy: {best_val_acc:.3f}')

```

```

pca=50, lr=0.001, hs=300, reg=0.0001, epochs=20 → val_acc=0.546
pca=50, lr=0.001, hs=300, reg=0.0001, epochs=25 → val_acc=0.545
pca=50, lr=0.001, hs=300, reg=1e-05, epochs=20 → val_acc=0.541
pca=50, lr=0.001, hs=300, reg=1e-05, epochs=25 → val_acc=0.547
pca=50, lr=0.001, hs=400, reg=0.0001, epochs=20 → val_acc=0.542
pca=50, lr=0.001, hs=400, reg=0.0001, epochs=25 → val_acc=0.542
pca=50, lr=0.001, hs=400, reg=1e-05, epochs=20 → val_acc=0.541
pca=50, lr=0.001, hs=400, reg=1e-05, epochs=25 → val_acc=0.534
pca=50, lr=0.001, hs=500, reg=0.0001, epochs=20 → val_acc=0.545
pca=50, lr=0.001, hs=500, reg=0.0001, epochs=25 → val_acc=0.542
pca=50, lr=0.001, hs=500, reg=1e-05, epochs=20 → val_acc=0.542
pca=50, lr=0.001, hs=500, reg=1e-05, epochs=25 → val_acc=0.553
pca=100, lr=0.001, hs=300, reg=0.0001, epochs=20 → val_acc=0.556
pca=100, lr=0.001, hs=300, reg=0.0001, epochs=25 → val_acc=0.563
pca=100, lr=0.001, hs=300, reg=1e-05, epochs=20 → val_acc=0.558
pca=100, lr=0.001, hs=300, reg=1e-05, epochs=25 → val_acc=0.551
pca=100, lr=0.001, hs=400, reg=0.0001, epochs=20 → val_acc=0.545
pca=100, lr=0.001, hs=400, reg=0.0001, epochs=25 → val_acc=0.560

```

```

pca=100, lr=0.001, hs=400, reg=1e-05, epochs=20 → val_acc=0.556
pca=100, lr=0.001, hs=400, reg=1e-05, epochs=25 → val_acc=0.559
pca=100, lr=0.001, hs=500, reg=0.0001, epochs=20 → val_acc=0.563
pca=100, lr=0.001, hs=500, reg=0.0001, epochs=25 → val_acc=0.563
pca=100, lr=0.001, hs=500, reg=1e-05, epochs=20 → val_acc=0.558
pca=100, lr=0.001, hs=500, reg=1e-05, epochs=25 → val_acc=0.574
pca=200, lr=0.001, hs=300, reg=0.0001, epochs=20 → val_acc=0.551
pca=200, lr=0.001, hs=300, reg=0.0001, epochs=25 → val_acc=0.552
pca=200, lr=0.001, hs=300, reg=1e-05, epochs=20 → val_acc=0.549
pca=200, lr=0.001, hs=300, reg=1e-05, epochs=25 → val_acc=0.552
pca=200, lr=0.001, hs=400, reg=0.0001, epochs=20 → val_acc=0.560
pca=200, lr=0.001, hs=400, reg=0.0001, epochs=25 → val_acc=0.573
pca=200, lr=0.001, hs=400, reg=1e-05, epochs=20 → val_acc=0.547
pca=200, lr=0.001, hs=400, reg=1e-05, epochs=25 → val_acc=0.563
pca=200, lr=0.001, hs=500, reg=0.0001, epochs=20 → val_acc=0.571
pca=200, lr=0.001, hs=500, reg=0.0001, epochs=25 → val_acc=0.556
pca=200, lr=0.001, hs=500, reg=1e-05, epochs=20 → val_acc=0.559
pca=200, lr=0.001, hs=500, reg=1e-05, epochs=25 → val_acc=0.563

```

best val accuracy: 0.574

12 Test your model!

Run your best model on the validation and test sets. You should achieve above 48% accuracy on the validation set and the test set.

```

[ ]: # transform val data through the same PCA
X_val_flat = data['X_val'].reshape(data['X_val'].shape[0], -1)
X_val_pca = best_pca.transform(X_val_flat)

y_val_pred = np.argmax(best_model.loss(X_val_pca), axis=1)
print('Validation set accuracy: ', (y_val_pred == data['y_val']).mean())

```

Validation set accuracy: 0.574

```

[ ]: # transform val data through the same PCA
X_test_flat = data['X_test'].reshape(data['X_test'].shape[0], -1)
X_test_pca = best_pca.transform(X_test_flat)

y_test_pred = np.argmax(best_model.loss(X_test_pca), axis=1)
print('Test set accuracy: ', (y_test_pred == data['y_test']).mean())

```

Test set accuracy: 0.546

```

[ ]: # Save best model
best_model.save("best_two_layer_net.npy")

```

best_two_layer_net.npy saved.

12.1 Inline Question 2:

Now that you have trained a Neural Network classifier, you may find that your testing accuracy is much lower than the training accuracy. In what ways can we decrease this gap? Select all that apply.

1. Train on a larger dataset.
2. Add more hidden units.
3. Increase the regularization strength.
4. None of the above.

Your Answer : 1, 3

Your Explanation :

Reasoning for 1 being a right answer - Increasing the training dataset size help decrease the variance as the model gets to generalize better as it sees more and more data in the training phase. As variance decreases (bias remains the same) thereby reducing the overall gap between testing and training accuracies.

Reasoning for 3 being a right answer - Increasing the regularization strength penalizes the weights from growing aggressively thereby avoiding the overfitting to the training data. Hence this helps with reducing the gap between training & testing accuracies.

Reasoning for 2 not being an answer - Increasing the hidden units increase the model complexity leading to greater variance and thereby causing larger gap between test and train accuracies.

[]:

features

April 17, 2026

Generative AI Use: For the purposes of the assignments, the use of generative AI is subject to the same policies regarding collaboration. Just as with other collaborators, each student must write down the solutions independently of the output of the interaction and the submission should include a note denoting the nature of the collaboration. The use of generative AI tools to substantially complete sections of the assignments is not in line with the spirit of the assignments, and would be a violation of the [Honor Code](#).

```
[1]: # This mounts your Google Drive to the Colab VM.
from google.colab import drive
drive.mount('/content/drive')

# TODO: Enter the foldername in your Drive where you have saved the unzipped
# assignment folder, e.g. 'cs231n/assignments/assignment1/'
FOLDERNAME = 'cs231n/assignments/assignment1/'
assert FOLDERNAME is not None, "[!] Enter the foldername."

# Now that we've mounted your Drive, this ensures that
# the Python interpreter of the Colab VM can load
# python files from within it.
import sys
sys.path.append('/content/drive/My Drive/{}'.format(FOLDERNAME))

# This downloads the CIFAR-10 dataset to your Drive
# if it doesn't already exist.
%cd /content/drive/My\ Drive/$FOLDERNAME/cs231n/datasets/
!bash get_datasets.sh
%cd /content/drive/My\ Drive/$FOLDERNAME
```

```
Mounted at /content/drive
/content/drive/My Drive/cs231n/assignments/assignment1/cs231n/datasets
/content/drive/My Drive/cs231n/assignments/assignment1
```

1 Image features exercise

Complete and hand in this completed worksheet (including its outputs and any supporting code outside of the worksheet) with your assignment submission. For more details see the [assignments page](#) on the course website.

We have seen that we can achieve reasonable performance on an image classification task by training

a linear classifier on the pixels of the input image. In this exercise we will show that we can improve our classification performance by training linear classifiers not on raw pixels but on features that are computed from the raw pixels.

All of your work for this exercise will be done in this notebook.

```
[2]: import random
import numpy as np
from cs231n.data_utils import load_CIFAR10
import matplotlib.pyplot as plt

%matplotlib inline
plt.rcParams['figure.figsize'] = (10.0, 8.0) # set default size of plots
plt.rcParams['image.interpolation'] = 'nearest'
plt.rcParams['image.cmap'] = 'gray'

# for auto-reloading external modules
# see http://stackoverflow.com/questions/1907993/
# ↪ autoreload-of-modules-in-ipython
%load_ext autoreload
%autoreload 2
```

1.1 Load data

Similar to previous exercises, we will load CIFAR-10 data from disk.

```
[3]: from cs231n.features import color_histogram_hsv, hog_feature

def get_CIFAR10_data(num_training=49000, num_validation=1000, num_test=1000):
    # Load the raw CIFAR-10 data
    cifar10_dir = 'cs231n/datasets/cifar-10-batches-py'

    # Cleaning up variables to prevent loading data multiple times (which may
    # ↪ cause memory issue)
    try:
        del X_train, y_train
        del X_test, y_test
        print('Clear previously loaded data.')
    except:
        pass

    X_train, y_train, X_test, y_test = load_CIFAR10(cifar10_dir)

    # Subsample the data
    mask = list(range(num_training, num_training + num_validation))
    X_val = X_train[mask]
    y_val = y_train[mask]
```



```

mask = list(range(num_training))
X_train = X_train[mask]
y_train = y_train[mask]
mask = list(range(num_test))
X_test = X_test[mask]
y_test = y_test[mask]

return X_train, y_train, X_val, y_val, X_test, y_test

X_train, y_train, X_val, y_val, X_test, y_test = get_CIFAR10_data()

```

1.2 Extract Features

For each image we will compute a Histogram of Oriented Gradients (HOG) as well as a color histogram using the hue channel in HSV color space. We form our final feature vector for each image by concatenating the HOG and color histogram feature vectors.

Roughly speaking, HOG should capture the texture of the image while ignoring color information, and the color histogram represents the color of the input image while ignoring texture. As a result, we expect that using both together ought to work better than using either alone. Verifying this assumption would be a good thing to try for your own interest.

The `hog_feature` and `color_histogram_hsv` functions both operate on a single image and return a feature vector for that image. The `extract_features` function takes a set of images and a list of feature functions and evaluates each feature function on each image, storing the results in a matrix where each column is the concatenation of all feature vectors for a single image.

```

[4]: from cs231n.features import *

# num_color_bins = 10 # Number of bins in the color histogram
num_color_bins = 25 # Number of bins in the color histogram
feature_fns = [hog_feature, lambda img: color_histogram_hsv(img,
    ↪nbin=num_color_bins)]
X_train_feats = extract_features(X_train, feature_fns, verbose=True)
X_val_feats = extract_features(X_val, feature_fns)
X_test_feats = extract_features(X_test, feature_fns)

# Preprocessing: Subtract the mean feature
mean_feat = np.mean(X_train_feats, axis=0, keepdims=True)
X_train_feats -= mean_feat
X_val_feats -= mean_feat
X_test_feats -= mean_feat

# Preprocessing: Divide by standard deviation. This ensures that each feature
# has roughly the same scale.
std_feat = np.std(X_train_feats, axis=0, keepdims=True)
X_train_feats /= std_feat
X_val_feats /= std_feat

```

```
X_test_feats /= std_feat
```

```
# Preprocessing: Add a bias dimension
```

```
X_train_feats = np.hstack([X_train_feats, np.ones((X_train_feats.shape[0], 1))])
```

```
X_val_feats = np.hstack([X_val_feats, np.ones((X_val_feats.shape[0], 1))])
```

```
X_test_feats = np.hstack([X_test_feats, np.ones((X_test_feats.shape[0], 1))])
```

```
Done extracting features for 1000 / 49000 images
Done extracting features for 2000 / 49000 images
Done extracting features for 3000 / 49000 images
Done extracting features for 4000 / 49000 images
Done extracting features for 5000 / 49000 images
Done extracting features for 6000 / 49000 images
Done extracting features for 7000 / 49000 images
Done extracting features for 8000 / 49000 images
Done extracting features for 9000 / 49000 images
Done extracting features for 10000 / 49000 images
Done extracting features for 11000 / 49000 images
Done extracting features for 12000 / 49000 images
Done extracting features for 13000 / 49000 images
Done extracting features for 14000 / 49000 images
Done extracting features for 15000 / 49000 images
Done extracting features for 16000 / 49000 images
Done extracting features for 17000 / 49000 images
Done extracting features for 18000 / 49000 images
Done extracting features for 19000 / 49000 images
Done extracting features for 20000 / 49000 images
Done extracting features for 21000 / 49000 images
Done extracting features for 22000 / 49000 images
Done extracting features for 23000 / 49000 images
Done extracting features for 24000 / 49000 images
Done extracting features for 25000 / 49000 images
Done extracting features for 26000 / 49000 images
Done extracting features for 27000 / 49000 images
Done extracting features for 28000 / 49000 images
Done extracting features for 29000 / 49000 images
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Done extracting features for 36000 / 49000 images
Done extracting features for 37000 / 49000 images
Done extracting features for 38000 / 49000 images
Done extracting features for 39000 / 49000 images
Done extracting features for 40000 / 49000 images
```

```

Done extracting features for 41000 / 49000 images
Done extracting features for 42000 / 49000 images
Done extracting features for 43000 / 49000 images
Done extracting features for 44000 / 49000 images
Done extracting features for 45000 / 49000 images
Done extracting features for 46000 / 49000 images
Done extracting features for 47000 / 49000 images
Done extracting features for 48000 / 49000 images
Done extracting features for 49000 / 49000 images

```

1.3 Train Softmax classifier on features

Using the Softmax code developed earlier in the assignment, train Softmax classifiers on top of the features extracted above; this should achieve better results than training them directly on top of raw pixels.

```

[5]: # Use the validation set to tune the learning rate and regularization strength

from cs231n.classifiers.linear_classifier import Softmax

learning_rates = [1e-7, 5e-8, 2e-7]
regularization_strengths = [5e4, 1e5, 5e5]

results = {}
best_val = -1
best_softmax = None

#####
# Done:
# Use the validation set to set the learning rate and regularization strength. #
# This should be identical to the validation that you did for the Softmax; save#
# the best trained classifier in best_softmax. If you carefully tune the model, #
# you should be able to get accuracy of above 0.42 on the validation set.      #
#####

for lr in learning_rates:
    for reg in regularization_strengths:
        # train softmax classifier
        softmax = Softmax()
        softmax.train(X_train_feats, y_train,
                      learning_rate=lr,
                      reg=reg,
                      num_iters=1500,
                      verbose=False)

        # training accuracy
        y_train_pred = softmax.predict(X_train_feats)
        train_accuracy = np.mean(y_train == y_train_pred)

```

```

# validation accuracy
y_val_pred = softmax.predict(X_val_feats)
val_accuracy = np.mean(y_val == y_val_pred)

results[(lr, reg)] = (train_accuracy, val_accuracy)

if val_accuracy > best_val:
    best_val = val_accuracy
    best_softmax = softmax

# Print out results.
for lr, reg in sorted(results):
    train_accuracy, val_accuracy = results[(lr, reg)]
    print('lr %e reg %e train accuracy: %f val accuracy: %f' % (
        lr, reg, train_accuracy, val_accuracy))

print('best validation accuracy achieved: %f' % best_val)

```

```

lr 5.000000e-08 reg 5.000000e+04 train accuracy: 0.290694 val accuracy: 0.306000
lr 5.000000e-08 reg 1.000000e+05 train accuracy: 0.422571 val accuracy: 0.428000
lr 5.000000e-08 reg 5.000000e+05 train accuracy: 0.415592 val accuracy: 0.401000
lr 1.000000e-07 reg 5.000000e+04 train accuracy: 0.420510 val accuracy: 0.428000
lr 1.000000e-07 reg 1.000000e+05 train accuracy: 0.416265 val accuracy: 0.407000
lr 1.000000e-07 reg 5.000000e+05 train accuracy: 0.417122 val accuracy: 0.428000
lr 2.000000e-07 reg 5.000000e+04 train accuracy: 0.423755 val accuracy: 0.425000
lr 2.000000e-07 reg 1.000000e+05 train accuracy: 0.422061 val accuracy: 0.426000
lr 2.000000e-07 reg 5.000000e+05 train accuracy: 0.403755 val accuracy: 0.417000
best validation accuracy achieved: 0.428000

```

```

[6]: # Evaluate your trained Softmax on the test set: you should be able to get at_
      ↪ least 0.42
y_test_pred = best_softmax.predict(X_test_feats)
test_accuracy = np.mean(y_test == y_test_pred)
print(test_accuracy)

```

0.42

```

[7]: # Save best softmax model
best_softmax.save("best_softmax_features.npy")

```

best_softmax_features.npy saved.

```

[8]: # An important way to gain intuition about how an algorithm works is to
# visualize the mistakes that it makes. In this visualization, we show examples
# of images that are misclassified by our current system. The first column
# shows images that our system labeled as "plane" but whose true label is

```

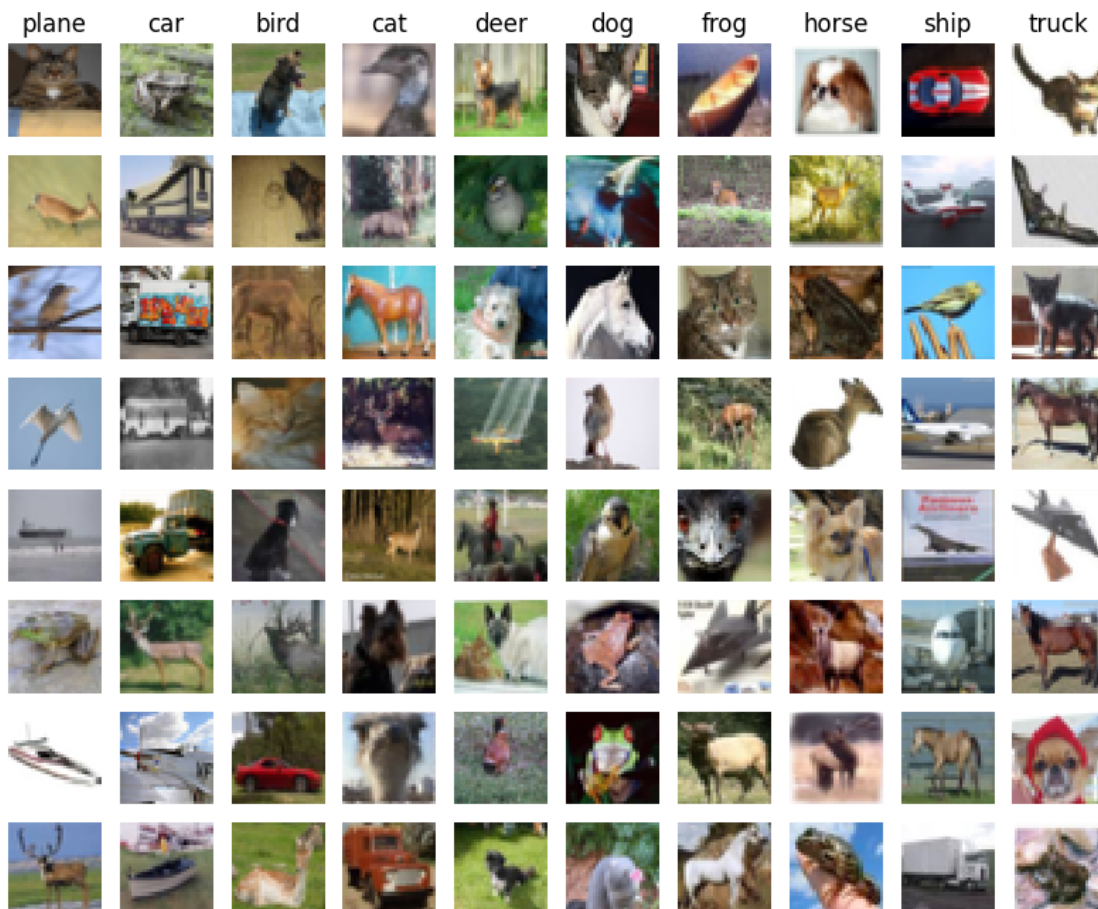
```

# something other than "plane".

examples_per_class = 8
classes = ['plane', 'car', 'bird', 'cat', 'deer', 'dog', 'frog', 'horse', '
↳'ship', 'truck']
for cls, cls_name in enumerate(classes):
    idxs = np.where((y_test != cls) & (y_test_pred == cls))[0]
    idxs = np.random.choice(idxs, examples_per_class, replace=False)
    for i, idx in enumerate(idxs):
        plt.subplot(examples_per_class, len(classes), i * len(classes) + cls +
↳1)

        plt.imshow(X_test[idx].astype('uint8'))
        plt.axis('off')
        if i == 0:
            plt.title(cls_name)
plt.show()

```



1.3.1 Inline question 1:

Describe the misclassification results that you see. Do they make sense?

Your Answer :

The misclassifications are mostly understandable. Our classifier relies on HOG and color histogram features, so it only “sees” texture and color and it is missing the semantic understanding of the underlying object or label.

A few examples that stand out: - Classifier misclassifies planes as ships and vice-versa because both tend to appear against blue backgrounds - Horses and deer look similar since they share the same brown tones and four-legged structure hence they get misclassified one as the other - Similarly, Red buses and red trucks look identical from a color histogram perspective - Cars and trucks share similar boxy edges, making them hard to tell apart from HOG alone.

Essentially the classifier is pattern-matching on surface-level cues rather than understanding objects.

1.4 Neural Network on image features

Earlier in this assignment we saw that training a two-layer neural network on raw pixels achieved better classification performance than linear classifiers on raw pixels. In this notebook we have seen that linear classifiers on image features outperform linear classifiers on raw pixels.

For completeness, we should also try training a neural network on image features. This approach should outperform all previous approaches: you should easily be able to achieve over 55% classification accuracy on the test set; our best model achieves about 60% classification accuracy.

```
[18]: # Preprocessing: Remove the bias dimension
      # Make sure to run this cell only ONCE
      print(X_train_feats.shape)
      X_train_feats = X_train_feats[:, :-1]
      X_val_feats = X_val_feats[:, :-1]
      X_test_feats = X_test_feats[:, :-1]
```

(49000, 167)

```
[24]: from cs231n.classifiers.fc_net import TwoLayerNet
      from cs231n.solver import Solver

      input_dim = X_train_feats.shape[1]
      num_classes = 10
      learning_rates = [1e-1]
      hidden_dims = [800, 1000]
      regs = [1e-4, 1e-5]
      num_epochs = [20, 25]

      data = {
          'X_train': X_train_feats,
          'y_train': y_train,
```

```

    'X_val': X_val_feats,
    'y_val': y_val,
    'X_test': X_test_feats,
    'y_test': y_test,
}

best_net = None
best_val_acc = 0
#####
# Done: Train a two-layer neural network on image features. You may want to #
# cross-validate various parameters as in previous sections. Store your best #
# model in the best_net variable. #
#####
for lr in learning_rates:
    for hd in hidden_dims:
        for reg in regs:
            for epochs in num_epochs:
                net = TwoLayerNet(input_dim, hd, num_classes, reg=reg)
                solver = Solver(net, data,
                                optim_config={'learning_rate': lr},
                                lr_decay=0.95,
                                num_epochs=epochs,
                                batch_size=200,
                                verbose=False,
                                )
                solver.train()

                val_acc = solver.best_val_acc
                print(f'lr={lr}, hd={hd}, reg={reg}, epochs={epochs} →
↪val_acc={val_acc:.3f}')

                if val_acc > best_val_acc:
                    best_val_acc = val_acc
                    best_net = net

print(f'\nbest val accuracy: {best_val_acc:.3f}')

```

```

lr=0.1, hd=800, reg=0.0001, epochs=20 → val_acc=0.601
lr=0.1, hd=800, reg=0.0001, epochs=25 → val_acc=0.612
lr=0.1, hd=800, reg=1e-05, epochs=20 → val_acc=0.605
lr=0.1, hd=800, reg=1e-05, epochs=25 → val_acc=0.609
lr=0.1, hd=1000, reg=0.0001, epochs=20 → val_acc=0.602
lr=0.1, hd=1000, reg=0.0001, epochs=25 → val_acc=0.608
lr=0.1, hd=1000, reg=1e-05, epochs=20 → val_acc=0.607
lr=0.1, hd=1000, reg=1e-05, epochs=25 → val_acc=0.616

```

best val accuracy: 0.616

```
[25]: # Run your best neural net classifier on the test set. You should be able  
# to get more than 58% accuracy. It is also possible to get >60% accuracy  
# with careful tuning.
```

```
y_test_pred = np.argmax(best_net.loss(data['X_test']), axis=1)  
test_acc = (y_test_pred == data['y_test']).mean()  
print(test_acc)
```

0.589

```
[26]: # Save best model  
best_net.save("best_two_layer_net_features.npy")
```

best_two_layer_net_features.npy saved.

```
[ ]:
```


FullyConnectedNets

April 17, 2026

Generative AI Use: For the purposes of the assignments, the use of generative AI is subject to the same policies regarding collaboration. Just as with other collaborators, each student must write down the solutions independently of the output of the interaction and the submission should include a note denoting the nature of the collaboration. The use of generative AI tools to substantially complete sections of the assignments is not in line with the spirit of the assignments, and would be a violation of the [Honor Code](#).

```
[1]: # This mounts your Google Drive to the Colab VM.
from google.colab import drive
drive.mount('/content/drive')

# TODO: Enter the foldername in your Drive where you have saved the unzipped
# assignment folder, e.g. 'cs231n/assignments/assignment1/'
FOLDERNAME = 'cs231n/assignments/assignment1/'
assert FOLDERNAME is not None, "[!] Enter the foldername."

# Now that we've mounted your Drive, this ensures that
# the Python interpreter of the Colab VM can load
# python files from within it.
import sys
sys.path.append('/content/drive/My Drive/{}'.format(FOLDERNAME))

# This downloads the CIFAR-10 dataset to your Drive
# if it doesn't already exist.
%cd /content/drive/My\ Drive/$FOLDERNAME/cs231n/datasets/
# %cd /content/drive/My\ Drive/$FOLDERNAME
!bash get_datasets.sh
%cd /content/drive/My\ Drive/$FOLDERNAME
```

```
Mounted at /content/drive
/content/drive/My Drive/cs231n/assignments/assignment1/cs231n/datasets
/content/drive/My Drive/cs231n/assignments/assignment1
```

1 Multi-Layer Fully Connected Network

In this exercise, you will implement a fully connected network with an arbitrary number of hidden layers.

```
[2]: # from google.colab import drive
# drive.mount('/content/drive')
```

Read through the FullyConnectedNet class in the file cs231n/classifiers/fc_net.py.

Implement the network initialization, forward pass, and backward pass. Throughout this assignment, you will be implementing layers in cs231n/layers.py. You can re-use your implementations for affine_forward, affine_backward, relu_forward, relu_backward, and softmax_loss from before. For right now, don't worry about implementing dropout or batch/layer normalization yet, as you will add those features later.

```
[3]: # Setup cell.
import time
import numpy as np
import matplotlib.pyplot as plt
from cs231n.classifiers.fc_net import *
from cs231n.data_utils import get_CIFAR10_data
from cs231n.gradient_check import eval_numerical_gradient, \
    eval_numerical_gradient_array
from cs231n.solver import Solver

%matplotlib inline
plt.rcParams["figure.figsize"] = (10.0, 8.0) # Set default size of plots.
plt.rcParams["image.interpolation"] = "nearest"
plt.rcParams["image.cmap"] = "gray"

%load_ext autoreload
%autoreload 2

def rel_error(x, y):
    """Returns relative error."""
    return np.max(np.abs(x - y) / (np.maximum(1e-8, np.abs(x) + np.abs(y))))
```

```
[4]: # Load the (preprocessed) CIFAR-10 data.
data = get_CIFAR10_data()
for k, v in list(data.items()):
    print(f"{k}: {v.shape}")
```

```
X_train: (49000, 3, 32, 32)
y_train: (49000,)
X_val: (1000, 3, 32, 32)
y_val: (1000,)
X_test: (1000, 3, 32, 32)
y_test: (1000,)
```

1.1 Initial Loss and Gradient Check

As a sanity check, run the following to check the initial loss and to gradient check the network both with and without regularization. This is a good way to see if the initial losses seem reasonable.

For gradient checking, you should expect to see errors around $1e-7$ or less.

```
[5]: np.random.seed(231)
N, D, H1, H2, C = 2, 15, 20, 30, 10
X = np.random.randn(N, D)
y = np.random.randint(C, size=(N,))

for reg in [0, 3.14]:
    print("Running check with reg = ", reg)
    model = FullyConnectedNet(
        [H1, H2],
        input_dim=D,
        num_classes=C,
        reg=reg,
        weight_scale=5e-2,
        dtype=np.float64
    )

    loss, grads = model.loss(X, y)
    print("Initial loss: ", loss)

    # Most of the errors should be on the order of e-7 or smaller.
    # NOTE: It is fine however to see an error for W2 on the order of e-5
    # for the check when reg = 0.0
    for name in sorted(grads):
        f = lambda _: model.loss(X, y)[0]
        grad_num = eval_numerical_gradient(f, model.params[name],
        verbose=False, h=1e-5)
        print(f"{name} relative error: {rel_error(grad_num, grads[name])}")
```

```
Running check with reg = 0
Initial loss: 2.300479089768492
W1 relative error: 1.0252674471656573e-07
W2 relative error: 2.2120479295080622e-05
W3 relative error: 4.5623278736665505e-07
b1 relative error: 4.6600944653202505e-09
b2 relative error: 2.085654276112763e-09
b3 relative error: 1.689724888469736e-10
Running check with reg = 3.14
Initial loss: 7.052114776533016
W1 relative error: 3.904541941902138e-09
W2 relative error: 6.86942277940646e-08
W3 relative error: 3.483989247437803e-08
b1 relative error: 1.4752427965311745e-08
```

```
b2 relative error: 1.4615869332918208e-09
b3 relative error: 1.3200479211447775e-10
```

As another sanity check, make sure your network can overfit on a small dataset of 50 images. First, we will try a three-layer network with 100 units in each hidden layer. In the following cell, tweak the **learning rate** and **weight initialization scale** to overfit and achieve 100% training accuracy within 20 epochs.

```
[8]: # TODO: Use a three-layer Net to overfit 50 training examples by  
# tweaking just the learning rate and initialization scale.
```

```
num_train = 50
small_data = {
    "X_train": data["X_train"][:num_train],
    "y_train": data["y_train"][:num_train],
    "X_val": data["X_val"],
    "y_val": data["y_val"],
}

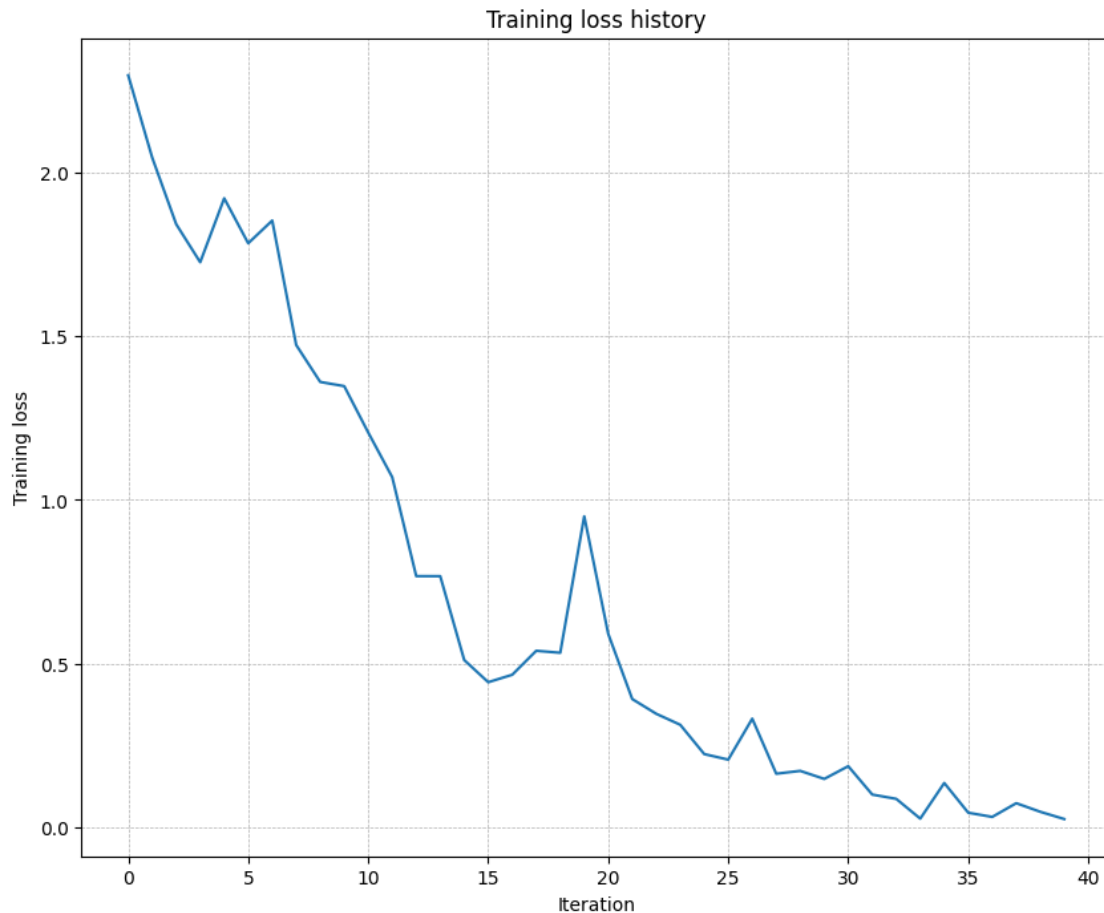
weight_scale = 1e-2
learning_rate = 1e-2

model = FullyConnectedNet(
    [100, 100],
    weight_scale=weight_scale,
    dtype=np.float64
)
solver = Solver(
    model,
    small_data,
    print_every=10,
    num_epochs=20,
    batch_size=25,
    update_rule="sgd",
    optim_config={"learning_rate": learning_rate},
)
solver.train()

plt.plot(solver.loss_history)
plt.title("Training loss history")
plt.xlabel("Iteration")
plt.ylabel("Training loss")
plt.grid(linestyle='--', linewidth=0.5)
plt.show()
```

```
(Iteration 1 / 40) loss: 2.295964
(Epoch 0 / 20) train acc: 0.220000; val_acc: 0.147000
(Epoch 1 / 20) train acc: 0.320000; val_acc: 0.153000
```

(Epoch 2 / 20) train acc: 0.400000; val_acc: 0.149000
(Epoch 3 / 20) train acc: 0.460000; val_acc: 0.150000
(Epoch 4 / 20) train acc: 0.600000; val_acc: 0.154000
(Epoch 5 / 20) train acc: 0.720000; val_acc: 0.171000
(Iteration 11 / 40) loss: 1.205734
(Epoch 6 / 20) train acc: 0.800000; val_acc: 0.147000
(Epoch 7 / 20) train acc: 0.760000; val_acc: 0.189000
(Epoch 8 / 20) train acc: 0.860000; val_acc: 0.178000
(Epoch 9 / 20) train acc: 0.880000; val_acc: 0.159000
(Epoch 10 / 20) train acc: 0.840000; val_acc: 0.164000
(Iteration 21 / 40) loss: 0.591158
(Epoch 11 / 20) train acc: 0.880000; val_acc: 0.179000
(Epoch 12 / 20) train acc: 0.940000; val_acc: 0.186000
(Epoch 13 / 20) train acc: 0.940000; val_acc: 0.164000
(Epoch 14 / 20) train acc: 0.980000; val_acc: 0.178000
(Epoch 15 / 20) train acc: 0.980000; val_acc: 0.189000
(Iteration 31 / 40) loss: 0.187313
(Epoch 16 / 20) train acc: 1.000000; val_acc: 0.185000
(Epoch 17 / 20) train acc: 1.000000; val_acc: 0.190000
(Epoch 18 / 20) train acc: 1.000000; val_acc: 0.194000
(Epoch 19 / 20) train acc: 1.000000; val_acc: 0.194000
(Epoch 20 / 20) train acc: 1.000000; val_acc: 0.198000



Now, try to use a five-layer network with 100 units on each layer to overfit on 50 training examples. Again, you will have to adjust the learning rate and weight initialization scale, but you should be able to achieve 100% training accuracy within 20 epochs.

[14]: *# TODO: Use a five-layer Net to overfit 50 training examples by
tweaking just the learning rate and initialization scale.*

```
num_train = 50
small_data = {
    'X_train': data['X_train'][:num_train],
    'y_train': data['y_train'][:num_train],
    'X_val': data['X_val'],
    'y_val': data['y_val'],
}

learning_rate = 1e-3 # Experiment with this!
weight_scale = 1e-1 # Experiment with this!
```

```

model = FullyConnectedNet(
    [100, 100, 100, 100],
    weight_scale=weight_scale,
    dtype=np.float64
)
solver = Solver(
    model,
    small_data,
    print_every=10,
    num_epochs=20,
    batch_size=25,
    update_rule='sgd',
    optim_config={'learning_rate': learning_rate},
)
solver.train()

plt.plot(solver.loss_history)
plt.title('Training loss history')
plt.xlabel('Iteration')
plt.ylabel('Training loss')
plt.grid(linestyle='--', linewidth=0.5)
plt.show()

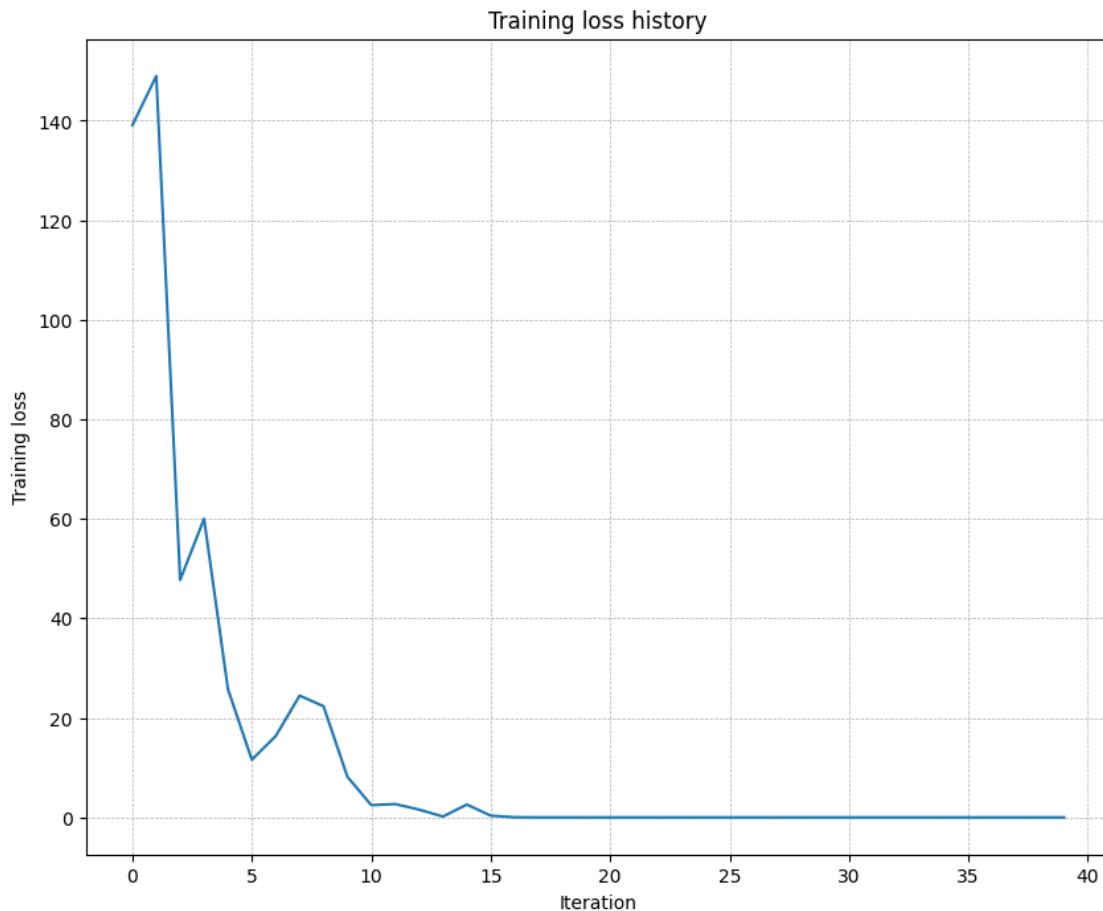
```

```

(Iteration 1 / 40) loss: 139.094248
(Epoch 0 / 20) train acc: 0.180000; val_acc: 0.123000
(Epoch 1 / 20) train acc: 0.320000; val_acc: 0.146000
(Epoch 2 / 20) train acc: 0.440000; val_acc: 0.127000
(Epoch 3 / 20) train acc: 0.480000; val_acc: 0.104000
(Epoch 4 / 20) train acc: 0.580000; val_acc: 0.132000
(Epoch 5 / 20) train acc: 0.720000; val_acc: 0.091000
(Iteration 11 / 40) loss: 2.462931
(Epoch 6 / 20) train acc: 0.940000; val_acc: 0.117000
(Epoch 7 / 20) train acc: 0.940000; val_acc: 0.116000
(Epoch 8 / 20) train acc: 0.960000; val_acc: 0.117000
(Epoch 9 / 20) train acc: 1.000000; val_acc: 0.118000
(Epoch 10 / 20) train acc: 1.000000; val_acc: 0.118000
(Iteration 21 / 40) loss: 0.000307
(Epoch 11 / 20) train acc: 1.000000; val_acc: 0.118000
(Epoch 12 / 20) train acc: 1.000000; val_acc: 0.118000
(Epoch 13 / 20) train acc: 1.000000; val_acc: 0.118000
(Epoch 14 / 20) train acc: 1.000000; val_acc: 0.118000
(Epoch 15 / 20) train acc: 1.000000; val_acc: 0.118000
(Iteration 31 / 40) loss: 0.000270
(Epoch 16 / 20) train acc: 1.000000; val_acc: 0.118000
(Epoch 17 / 20) train acc: 1.000000; val_acc: 0.118000
(Epoch 18 / 20) train acc: 1.000000; val_acc: 0.118000
(Epoch 19 / 20) train acc: 1.000000; val_acc: 0.118000

```

(Epoch 20 / 20) train acc: 1.000000; val_acc: 0.118000



1.2 Inline Question 1:

Did you notice anything about the comparative difficulty of training the three-layer network vs. training the five-layer network? In particular, based on your experience, which network seemed more sensitive to the initialization scale? Why do you think that is the case?

1.3 Answer:

Yes, the five-layer network was significantly harder to train than the three-layer network.

The three-layer network converged to 100% training accuracy easily with `weight_scale=1e-2` and `learning_rate=1e-2`. The five-layer network completely failed with the same settings, loss barely moved from 2.302.

The five-layer network was far more sensitive to initialization scale. With `weight_scale=1e-2` the five-layer network suffered from vanishing gradients, i.e, signals shrank at each layer and by the time they reached the early layers, they were too small to drive meaningful learning. Only by

increasing `weight_scale` to `1e-1` did the network finally learn, reaching 100% training accuracy by epoch 9.

This is because each layer computes $X @ W$ where $W \approx 0.01$, so activations shrink by a factor of 0.01 at every layer. With 5 layers this compounds to $0.01^5 = 10^{-10}$, essentially zero. Gradients flowing backward shrink the same way, meaning early weights receive near-zero updates and never learn. A 3-layer network only compounds 3 times ($\sim 0.01^3 = 10^{-6}$), still small but manageable enough to learn. This is precisely the motivation behind techniques like batch normalization and Xavier/He initialization to avoid vanishing/exploding gradients problems.

2 Update rules

So far we have used vanilla stochastic gradient descent (SGD) as our update rule. More sophisticated update rules can make it easier to train deep networks. We will implement a few of the most commonly used update rules and compare them to vanilla SGD.

2.1 SGD+Momentum

Stochastic gradient descent with momentum is a widely used update rule that tends to make deep networks converge faster than vanilla stochastic gradient descent. See the Momentum Update section at <http://cs231n.github.io/neural-networks-3/#sgd> for more information.

Open the file `cs231n/optim.py` and read the documentation at the top of the file to make sure you understand the API. Implement the SGD+momentum update rule in the function `sgd_momentum` and run the following to check your implementation. You should see errors less than $e-8$.

```
[16]: from cs231n.optim import sgd_momentum

N, D = 4, 5
w = np.linspace(-0.4, 0.6, num=N*D).reshape(N, D)
dw = np.linspace(-0.6, 0.4, num=N*D).reshape(N, D)
v = np.linspace(0.6, 0.9, num=N*D).reshape(N, D)

config = {"learning_rate": 1e-3, "velocity": v}
next_w, _ = sgd_momentum(w, dw, config=config)

expected_next_w = np.asarray([
    [ 0.1406,      0.20738947,  0.27417895,  0.34096842,  0.40775789],
    [ 0.47454737,  0.54133684,  0.60812632,  0.67491579,  0.74170526],
    [ 0.80849474,  0.87528421,  0.94207368,  1.00886316,  1.07565263],
    [ 1.14244211,  1.20923158,  1.27602105,  1.34281053,  1.4096      ]])
expected_velocity = np.asarray([
    [ 0.5406,      0.55475789,  0.56891579,  0.58307368,  0.59723158],
    [ 0.61138947,  0.62554737,  0.63970526,  0.65386316,  0.66802105],
    [ 0.68217895,  0.69633684,  0.71049474,  0.72465263,  0.73881053],
    [ 0.75296842,  0.76712632,  0.78128421,  0.79544211,  0.8096      ]])

# Should see relative errors around e-8 or less
print("next_w error: ", rel_error(next_w, expected_next_w))
```

```
print("velocity error: ", rel_error(expected_velocity, config["velocity"]))
```

next_w error: 8.882347033505819e-09

velocity error: 4.269287743278663e-09

Once you have done so, run the following to train a six-layer network with both SGD and SGD+momentum. You should see the SGD+momentum update rule converge faster.

```
[17]: num_train = 4000
      small_data = {
          'X_train': data['X_train'][:num_train],
          'y_train': data['y_train'][:num_train],
          'X_val': data['X_val'],
          'y_val': data['y_val'],
      }

      solvers = {}

      for update_rule in ['sgd', 'sgd_momentum']:
          print('Running with ', update_rule)
          model = FullyConnectedNet(
              [100, 100, 100, 100, 100],
              weight_scale=5e-2
          )

          solver = Solver(
              model,
              small_data,
              num_epochs=5,
              batch_size=100,
              update_rule=update_rule,
              optim_config={'learning_rate': 5e-3},
              verbose=True,
          )
          solvers[update_rule] = solver
          solver.train()

      fig, axes = plt.subplots(3, 1, figsize=(15, 15))

      axes[0].set_title('Training loss')
      axes[0].set_xlabel('Iteration')
      axes[1].set_title('Training accuracy')
      axes[1].set_xlabel('Epoch')
      axes[2].set_title('Validation accuracy')
      axes[2].set_xlabel('Epoch')

      for update_rule, solver in solvers.items():
          axes[0].plot(solver.loss_history, label=f"loss_{update_rule}")
```

```

axes[1].plot(solver.train_acc_history, label=f"train_acc_{update_rule}")
axes[2].plot(solver.val_acc_history, label=f"val_acc_{update_rule}")

for ax in axes:
    ax.legend(loc="best", ncol=4)
    ax.grid(linestyle='--', linewidth=0.5)

plt.show()

```

Running with `sgd`

```

(Iteration 1 / 200) loss: 2.475477
(Epoch 0 / 5) train acc: 0.110000; val_acc: 0.113000
(Iteration 11 / 200) loss: 2.341384
(Iteration 21 / 200) loss: 2.289701
(Iteration 31 / 200) loss: 2.177896
(Epoch 1 / 5) train acc: 0.229000; val_acc: 0.201000
(Iteration 41 / 200) loss: 2.154686
(Iteration 51 / 200) loss: 1.984470
(Iteration 61 / 200) loss: 2.031636
(Iteration 71 / 200) loss: 2.083912
(Epoch 2 / 5) train acc: 0.248000; val_acc: 0.257000
(Iteration 81 / 200) loss: 1.937931
(Iteration 91 / 200) loss: 2.058391
(Iteration 101 / 200) loss: 1.872363
(Iteration 111 / 200) loss: 1.912813
(Epoch 3 / 5) train acc: 0.317000; val_acc: 0.284000
(Iteration 121 / 200) loss: 1.815903
(Iteration 131 / 200) loss: 1.970755
(Iteration 141 / 200) loss: 1.802829
(Iteration 151 / 200) loss: 1.766025
(Epoch 4 / 5) train acc: 0.365000; val_acc: 0.311000
(Iteration 161 / 200) loss: 1.839869
(Iteration 171 / 200) loss: 1.860592
(Iteration 181 / 200) loss: 1.806990
(Iteration 191 / 200) loss: 1.663018
(Epoch 5 / 5) train acc: 0.372000; val_acc: 0.324000

```

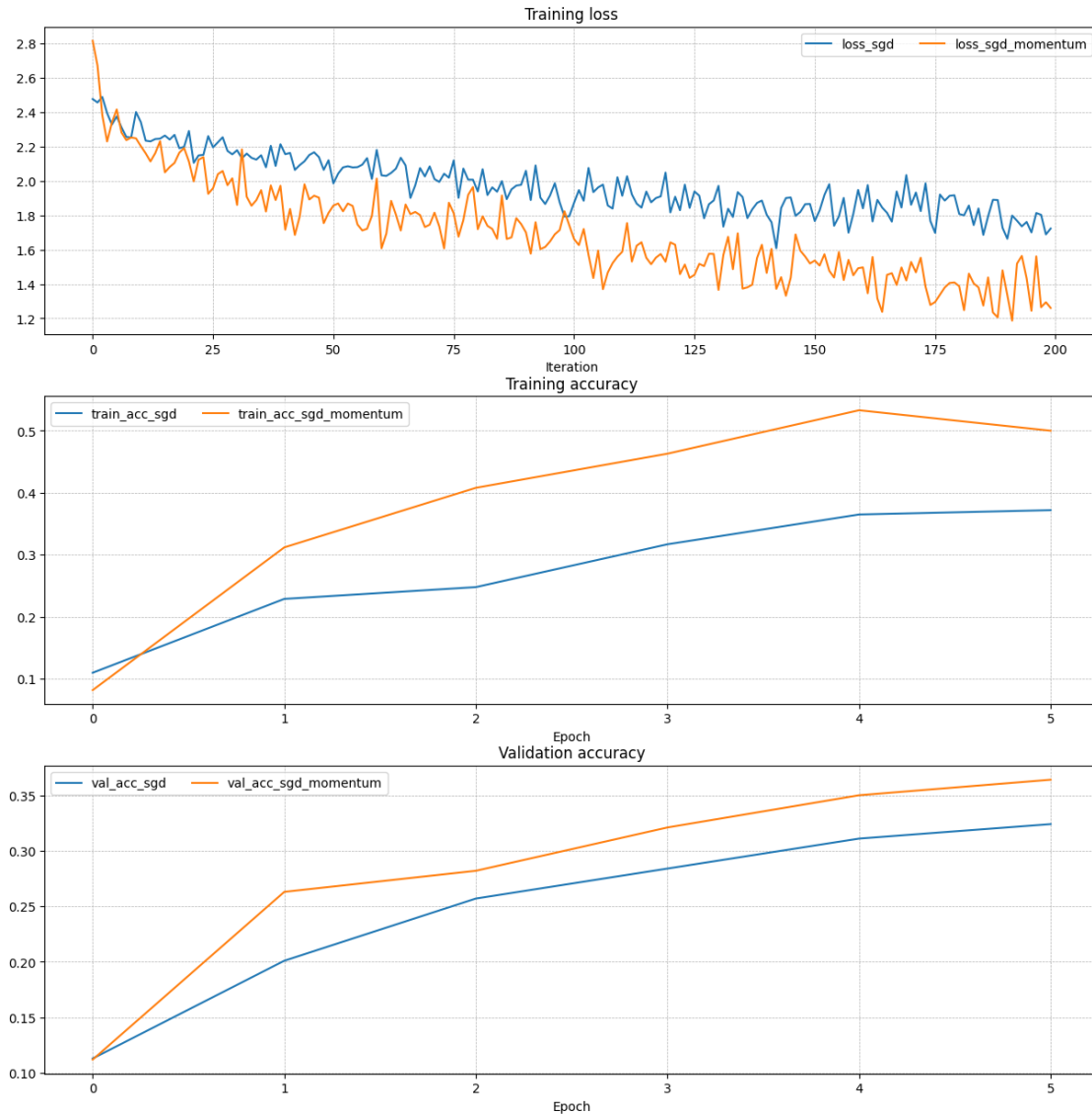
Running with `sgd_momentum`

```

(Iteration 1 / 200) loss: 2.814150
(Epoch 0 / 5) train acc: 0.082000; val_acc: 0.112000
(Iteration 11 / 200) loss: 2.201798
(Iteration 21 / 200) loss: 2.111939
(Iteration 31 / 200) loss: 1.859592
(Epoch 1 / 5) train acc: 0.312000; val_acc: 0.263000
(Iteration 41 / 200) loss: 1.715869
(Iteration 51 / 200) loss: 1.855734
(Iteration 61 / 200) loss: 1.607224
(Iteration 71 / 200) loss: 1.744718

```

(Epoch 2 / 5) train acc: 0.408000; val_acc: 0.282000
(Iteration 81 / 200) loss: 1.718483
(Iteration 91 / 200) loss: 1.700143
(Iteration 101 / 200) loss: 1.661040
(Iteration 111 / 200) loss: 1.587237
(Epoch 3 / 5) train acc: 0.463000; val_acc: 0.321000
(Iteration 121 / 200) loss: 1.642048
(Iteration 131 / 200) loss: 1.365543
(Iteration 141 / 200) loss: 1.464201
(Iteration 151 / 200) loss: 1.536307
(Epoch 4 / 5) train acc: 0.533000; val_acc: 0.350000
(Iteration 161 / 200) loss: 1.496579
(Iteration 171 / 200) loss: 1.528120
(Iteration 181 / 200) loss: 1.386769
(Iteration 191 / 200) loss: 1.334753
(Epoch 5 / 5) train acc: 0.500000; val_acc: 0.364000



2.2 RMSProp and Adam

RMSProp [1] and Adam [2] are update rules that set per-parameter learning rates by using a running average of the second moments of gradients.

In the file `cs231n/optim.py`, implement the RMSProp update rule in the `rmsprop` function and implement the Adam update rule in the `adam` function, and check your implementations using the tests below.

NOTE: Please implement the *complete* Adam update rule (with the bias correction mechanism), not the first simplified version mentioned in the course notes.

[1] Tijmen Tieleman and Geoffrey Hinton. “Lecture 6.5-rmsprop: Divide the gradient by a running average of its recent magnitude.” COURSE: Neural Networks for Machine Learning 4 (2012).

[2] Diederik Kingma and Jimmy Ba, “Adam: A Method for Stochastic Optimization”, ICLR 2015.

```
[18]: # Test RMSProp implementation
from cs231n.optim import rmsprop

N, D = 4, 5
w = np.linspace(-0.4, 0.6, num=N*D).reshape(N, D)
dw = np.linspace(-0.6, 0.4, num=N*D).reshape(N, D)
cache = np.linspace(0.6, 0.9, num=N*D).reshape(N, D)

config = {'learning_rate': 1e-2, 'cache': cache}
next_w, _ = rmsprop(w, dw, config=config)

expected_next_w = np.asarray([
    [-0.39223849, -0.34037513, -0.28849239, -0.23659121, -0.18467247],
    [-0.132737, -0.08078555, -0.02881884, 0.02316247, 0.07515774],
    [0.12716641, 0.17918792, 0.23122175, 0.28326742, 0.33532447],
    [0.38739248, 0.43947102, 0.49155973, 0.54365823, 0.59576619]])
expected_cache = np.asarray([
    [0.5976, 0.6126277, 0.6277108, 0.64284931, 0.65804321],
    [0.67329252, 0.68859723, 0.70395734, 0.71937285, 0.73484377],
    [0.75037008, 0.7659518, 0.78158892, 0.79728144, 0.81302936],
    [0.82883269, 0.84469141, 0.86060554, 0.87657507, 0.8926]])

# You should see relative errors around e-7 or less
print('next_w error: ', rel_error(expected_next_w, next_w))
print('cache error: ', rel_error(expected_cache, config['cache']))
```

next_w error: 9.524687511038133e-08

cache error: 2.6477955807156126e-09

```
[19]: # Test Adam implementation
from cs231n.optim import adam

N, D = 4, 5
w = np.linspace(-0.4, 0.6, num=N*D).reshape(N, D)
dw = np.linspace(-0.6, 0.4, num=N*D).reshape(N, D)
m = np.linspace(0.6, 0.9, num=N*D).reshape(N, D)
v = np.linspace(0.7, 0.5, num=N*D).reshape(N, D)

config = {'learning_rate': 1e-2, 'm': m, 'v': v, 't': 5}
next_w, _ = adam(w, dw, config=config)

expected_next_w = np.asarray([
    [-0.40094747, -0.34836187, -0.29577703, -0.24319299, -0.19060977],
    [-0.1380274, -0.08544591, -0.03286534, 0.01971428, 0.0722929],
    [0.1248705, 0.17744702, 0.23002243, 0.28259667, 0.33516969],
    [0.38774145, 0.44031188, 0.49288093, 0.54544852, 0.59801459]])
```

```

expected_v = np.asarray([
    [ 0.69966,      0.68908382,   0.67851319,   0.66794809,   0.65738853,],
    [ 0.64683452,   0.63628604,   0.6257431,    0.61520571,   0.60467385,],
    [ 0.59414753,   0.58362676,   0.57311152,   0.56260183,   0.55209767,],
    [ 0.54159906,   0.53110598,   0.52061845,   0.51013645,   0.49966,   ]])
expected_m = np.asarray([
    [ 0.48,          0.49947368,   0.51894737,   0.53842105,   0.55789474],
    [ 0.57736842,   0.59684211,   0.61631579,   0.63578947,   0.65526316],
    [ 0.67473684,   0.69421053,   0.71368421,   0.73315789,   0.75263158],
    [ 0.77210526,   0.79157895,   0.81105263,   0.83052632,   0.85         ]])

# You should see relative errors around e-7 or less
print('next_w error: ', rel_error(expected_next_w, next_w))
print('v error: ', rel_error(expected_v, config['v']))
print('m error: ', rel_error(expected_m, config['m']))

```

```

next_w error:  1.1395691798535431e-07
v error:  4.208314038113071e-09
m error:  4.214963193114416e-09

```

Once you have debugged your RMSProp and Adam implementations, run the following to train a pair of deep networks using these new update rules:

Add blockquote

```

[20]: learning_rates = {'rmsprop': 1e-4, 'adam': 1e-3}
for update_rule in ['adam', 'rmsprop']:
    print('Running with ', update_rule)
    model = FullyConnectedNet(
        [100, 100, 100, 100, 100],
        weight_scale=5e-2
    )
    solver = Solver(
        model,
        small_data,
        num_epochs=5,
        batch_size=100,
        update_rule=update_rule,
        optim_config={'learning_rate': learning_rates[update_rule]},
        verbose=True
    )
    solvers[update_rule] = solver
    solver.train()
    print()

fig, axes = plt.subplots(3, 1, figsize=(15, 15))

axes[0].set_title('Training loss')

```

```

axes[0].set_xlabel('Iteration')
axes[1].set_title('Training accuracy')
axes[1].set_xlabel('Epoch')
axes[2].set_title('Validation accuracy')
axes[2].set_xlabel('Epoch')

for update_rule, solver in solvers.items():
    axes[0].plot(solver.loss_history, label=f"{update_rule}")
    axes[1].plot(solver.train_acc_history, label=f"{update_rule}")
    axes[2].plot(solver.val_acc_history, label=f"{update_rule}")

for ax in axes:
    ax.legend(loc='best', ncol=4)
    ax.grid(linestyle='--', linewidth=0.5)

plt.show()

```

Running with adam

```

(Iteration 1 / 200) loss: 2.534728
(Epoch 0 / 5) train acc: 0.123000; val_acc: 0.129000
(Iteration 11 / 200) loss: 2.121538
(Iteration 21 / 200) loss: 1.976581
(Iteration 31 / 200) loss: 1.855805
(Epoch 1 / 5) train acc: 0.366000; val_acc: 0.316000
(Iteration 41 / 200) loss: 1.440984
(Iteration 51 / 200) loss: 1.833658
(Iteration 61 / 200) loss: 1.804008
(Iteration 71 / 200) loss: 1.619363
(Epoch 2 / 5) train acc: 0.458000; val_acc: 0.375000
(Iteration 81 / 200) loss: 1.583346
(Iteration 91 / 200) loss: 1.428342
(Iteration 101 / 200) loss: 1.578038
(Iteration 111 / 200) loss: 1.631269
(Epoch 3 / 5) train acc: 0.460000; val_acc: 0.348000
(Iteration 121 / 200) loss: 1.607584
(Iteration 131 / 200) loss: 1.509495
(Iteration 141 / 200) loss: 1.408251
(Iteration 151 / 200) loss: 1.179472
(Epoch 4 / 5) train acc: 0.557000; val_acc: 0.377000
(Iteration 161 / 200) loss: 1.358903
(Iteration 171 / 200) loss: 1.180278
(Iteration 181 / 200) loss: 1.064116
(Iteration 191 / 200) loss: 1.149457
(Epoch 5 / 5) train acc: 0.593000; val_acc: 0.382000

```

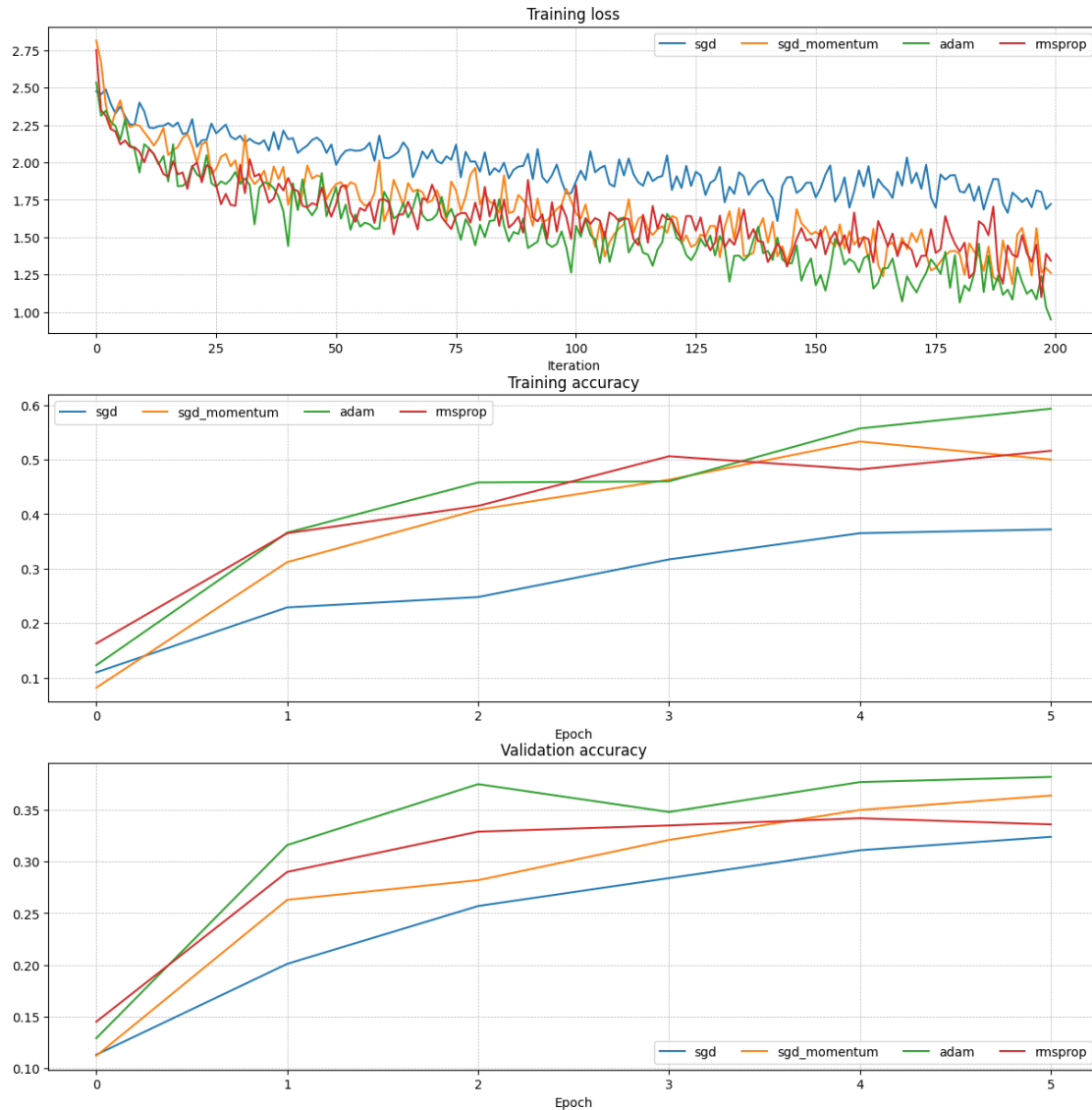
Running with rmsprop

```

(Iteration 1 / 200) loss: 2.751459

```


(Epoch 0 / 5) train acc: 0.163000; val_acc: 0.145000
(Iteration 11 / 200) loss: 2.000681
(Iteration 21 / 200) loss: 1.972926
(Iteration 31 / 200) loss: 1.984343
(Epoch 1 / 5) train acc: 0.365000; val_acc: 0.290000
(Iteration 41 / 200) loss: 1.896108
(Iteration 51 / 200) loss: 1.722322
(Iteration 61 / 200) loss: 1.753068
(Iteration 71 / 200) loss: 1.850992
(Epoch 2 / 5) train acc: 0.415000; val_acc: 0.329000
(Iteration 81 / 200) loss: 1.613536
(Iteration 91 / 200) loss: 1.883486
(Iteration 101 / 200) loss: 1.847562
(Iteration 111 / 200) loss: 1.624586
(Epoch 3 / 5) train acc: 0.506000; val_acc: 0.335000
(Iteration 121 / 200) loss: 1.619152
(Iteration 131 / 200) loss: 1.645099
(Iteration 141 / 200) loss: 1.334505
(Iteration 151 / 200) loss: 1.428335
(Epoch 4 / 5) train acc: 0.482000; val_acc: 0.342000
(Iteration 161 / 200) loss: 1.501642
(Iteration 171 / 200) loss: 1.413864
(Iteration 181 / 200) loss: 1.404305
(Iteration 191 / 200) loss: 1.445523
(Epoch 5 / 5) train acc: 0.516000; val_acc: 0.336000



2.3 Inline Question 2:

AdaGrad, like Adam, is a per-parameter optimization method that uses the following update rule:

```
cache += dw**2
w += - learning_rate * dw / (np.sqrt(cache) + eps)
```

John notices that when he was training a network with AdaGrad that the updates became very small, and that his network was learning slowly. Using your knowledge of the AdaGrad update rule, why do you think the updates would become very small? Would Adam have the same issue?

2.4 Answer:

The problem with AdaGrad is that cache just keeps growing as every iteration adds more squared gradients and nothing ever brings it back down. So over time the denominator np.sqrt(cache) gets larger and larger, which means the effective learning rate keeps shrinking. Eventually updates become so tiny the network basically stops learning, even if the gradients themselves are still reasonable.

```
cache += dw**2
```

Adam doesn't have this problem because instead of accumulating gradients forever, it uses an exponential moving average:

```
v = beta2 * v + (1 - beta2) * dw**2
```

3 Train a Good Model!

Train the best fully connected model that you can on CIFAR-10, storing your best model in the `best_model` variable. We require you to get at least 50% accuracy on the validation set using a fully connected network.

If you are careful it should be possible to get accuracies above 55%, but we don't require it for this part and won't assign extra credit for doing so. Later in the next assignment, we will ask you to train the best convolutional network that you can on CIFAR-10, and we would prefer that you spend your effort working on convolutional networks rather than fully connected networks.

Note: In the next assignment, you will learn techniques like BatchNormalization and Dropout which can help you train powerful models.

```
[22]: best_model = None

#####
# Done: Train the best FullyConnectedNet that you can on CIFAR-10. You might #
# find batch/layer normalization and dropout useful. Store your best model in #
# the best_model variable.                                                    #
#####
# *****START OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
best_val_acc = 0

configs = [
    {'hidden': [500, 500],      'lr': 1e-3, 'reg': 1e-4, 'ws': 5e-2},
    {'hidden': [500, 500, 500], 'lr': 1e-3, 'reg': 1e-4, 'ws': 5e-2},
    {'hidden': [1000, 500],     'lr': 1e-3, 'reg': 1e-4, 'ws': 5e-2},
    {'hidden': [1000, 1000],    'lr': 1e-3, 'reg': 1e-5, 'ws': 5e-2},
]

for cfg in configs:
    model = FullyConnectedNet(
        cfg['hidden'],
        weight_scale=cfg['ws'],
```

```

        reg=cfg['reg'],
    )
    solver = Solver(
        model,
        data,
        num_epochs=20,
        batch_size=100,
        update_rule='adam',
        optim_config={'learning_rate': cfg['lr']},
        lr_decay=0.95,
        verbose=False,
    )
    solver.train()
    val_acc = solver.best_val_acc
    print(f"hidden={cfg['hidden']} → val_acc={val_acc:.3f}")

    if val_acc > best_val_acc:
        best_val_acc = val_acc
        best_model = model

print(f'\nbest val accuracy: {best_val_acc:.3f}')

# *****END OF YOUR CODE (DO NOT DELETE/MODIFY THIS LINE)*****
#####
#                                     END OF YOUR CODE                                     #
#####

```

```

hidden=[500, 500] → val_acc=0.500
hidden=[500, 500, 500] → val_acc=0.512
hidden=[1000, 500] → val_acc=0.463
hidden=[1000, 1000] → val_acc=0.436

```

```
best val accuracy: 0.512
```

4 Test Your Model!

Run your best model on the validation and test sets. You should achieve at least 50% accuracy on the validation set and the test set.

```

[23]: y_test_pred = np.argmax(best_model.loss(data['X_test']), axis=1)
      y_val_pred = np.argmax(best_model.loss(data['X_val']), axis=1)
      print('Validation set accuracy: ', (y_val_pred == data['y_val']).mean())
      print('Test set accuracy: ', (y_test_pred == data['y_test']).mean())

```

```

Validation set accuracy: 0.512
Test set accuracy: 0.508

```